Privacy-Preserving Vertical Federated Regression

with Threshold Decryption and Differentially Private Releases

Working draft — protocol v3.1, honest-but-curious model

October 24, 2026

Abstract

We study two-party vertical federated least squares between a researcher R, who holds a cohort with features $\mathbf{X}_R$, and an organization O (for example a national statistical agency), which holds a register with features $\mathbf{X}_O$ and the response $\mathbf{y}$. Records are linked by labeled private set intersection with an oblivious pseudorandom function: R learns which of its cohort members are in O's register and where, and O learns only public sizes. The statistics that depend on O's data are aggregated under two-party threshold CKKS. The secret key is split additively between the parties, so neither can decrypt alone, and O never decrypts at all. Instead, O adds the differential-privacy noise inside its share of each decryption, as an exact discrete Gaussian on the coefficient lattice that also floods the encryption error. R therefore receives only noisy statistics, and O can enforce a privacy budget by withholding its share. R's own Gram block is computed locally and is never noised. Each run makes two releases. The first, of the Gram blocks only, chooses the ridge. The second, whose noise is independent of that choice, gives the estimate, a Wald interval for the ridge target from a parametric bootstrap of the mechanism, and a bias-aware interval for the regression coefficients. We give the exact replace-one sensitivity, zCDP accounting with a tight conversion, simulation-based output privacy against an honest-but-curious researcher, and the statistical guarantees of the two-stage procedure. In a simulation of the release and solve steps the Wald intervals cover 94.4–96.2% (target 95.5%) with and without the ridge. The accuracy results so far use an earlier single-release mechanism. On real data, prediction comes within 10% of non-private test error at 1.4k–11k matched records for $\varepsilon \in \{1, 2\}$, and coefficient-level accuracy needs roughly ten times more data. End-to-end runs of a full OpenFHE implementation on the protein dataset (19,206 matched records) reproduce a plaintext simulation of the mechanism and, after each party normalizes its own block of every row, come within 2–6% of non-private test error at $\varepsilon \in \{1, 4\}$. [**Check 1:** The accuracy numbers in this abstract come from the earlier single-release mechanism. Replace them after the reruns under v3.1 (two releases, ridge from stage 1, delta = 1e-9, bias correction off).]

Contents

1 Introduction **6**

2 Related work **7**

3 Setting and notation **8**

4 Threat model and guarantees **9**

5 Building blocks **10**
5.1 Two-party threshold CKKS . . . 10
5.2 Oblivious PRF . . . 11
5.3 Labeled PSI . . . 11
5.4 zCDP and the discrete Gaussian . . . 11

6 The protocol **11**
6.1 Per-party preprocessing . . . 15

7 Output privacy **20**
7.1 Sensitivity under replace-one adjacency . . . 20
7.2 The release mechanism . . . 21
7.3 Calibration . . . 23
7.4 The ledger and composition across runs . . . 25

8 Input privacy, cohort privacy and correctness **26**

9 The statistical procedure **28**
9.1 Estimands . . . 28
9.2 Stage 1: choosing the ridge . . . 28
9.3 Stage 2: estimate and gate . . . 29
9.4 Intervals . . . 29
9.5 When can the ridge be dropped? . . . 30
9.6 Status of the statistical guarantees . . . 33

10 Mechanism bias and the optional correction **33**
10.1 Noise model . . . 33
10.2 Result . . . 34
10.3 Schur-complement reading . . . 34
10.4 Ridge . . . 34
10.5 Why the ridge must not depend on the estimation noise . . . 35
10.6 Optional zero-budget correction . . . 35

11 Prediction after training **35**

12 Parameters **36**

13 Empirical evaluation **39**
13.1 The cryptographic layer is exact . . . 39
13.2 Calibration under v3.1 . . . 40
13.3 Interval coverage under v3.1 . . . 41
13.4 End-to-end runs of the implementation . . . 41
13.5 Coefficient accuracy against intersection size (earlier mechanism) . . . 42
13.6 Does the bias correction matter? (earlier mechanism) . . . 44
13.7 Benchmark against AdaSSP (earlier mechanism) . . . 44
13.8 Intersection size on real data (earlier mechanism) . . . 46

13.9 Where the utility goes . . . 46

14 Cost **48**

15 Extensions and limitations **49**

16 Summary and open items **50**

A Proof of Theorem 10.1 **50**

B Labeled PSI with an OPRF and record alignment **52**

C Status of the proofs **53**

Items to check

1 The accuracy numbers in this abstract come from the earlier single-release mechanism. Replace them after the reruns under v3.1 (two releases, ridge from stage 1, delta = 1e-9, bias correction off). . . . 1

2 All bibliographic details in this section were written from memory and must be verified. The positioning against Froelicher et al. also needs a careful read of their papers. . . . 7

3 ChooseBounds: the evaluation uses the 99.5% quantile of each block's norm on the preprocessed reference data. The optimized choice (quantile grid minimizing first-order MSE plus squared clipping bias) has not been specified or evaluated. . . . 11

6 Preprocessing defaults: which transform to recommend for prediction and for interpretation; the appended log-norm feature and winsorizing y are not yet evaluated. 16

4 Replace the bisection for lambda-O by the closed form of the proofs (Lemma S3(c)). 18

5 U-yy as corrected by the proof review (the specification spent alpha-U twice and left sigma-yy undefined). Confirm. . . . 18

7 DeltaRep in regime 3: the specification says one-dimensional maximization rounded up; the proofs give an exact enumeration over the roots of three cubics. Adopt the exact enumeration (keep rounding up as a safeguard)? . . . 20

8 Design choice raised by the remark above: release the full matrix B (both triangles, independent noise) instead of its upper triangle? Same worst-case privacy cost, lower variance off the diagonal; the bias formulas and the bootstrap would need the corresponding noise covariance. . . . 21

9 Composition: the adaptive hybrid of Proposition 7.2 is argued here but not yet written out in the proofs, which prove the non-adaptive case. The literature was checked only through search-engine extracts, not the full texts: Feldman and Zrnic's Theorem 1.1 as quoted from the arXiv version (check its number in the NeurIPS version, and the page numbers); the pages of Rogers et al. (1921–1929); the volume and pages of Whitehouse et al. (PMLR 202, 36990–37007). . . . 26

10 G4: derive a worst-case, data-independent bound on the BFV evaluation noise of the PSI replies after the switch to q', so that B-psi is proven (the implementation's value is measured). The circuit-privacy statements of CLR17 and the role of OPRF preprocessing in CHLR18 were checked only through search-engine extracts; confirm against the full texts, including whether CHLR18 state that OPRF preprocessing removes the need for circuit privacy. 27
11 Choose the definition of beta (working model on clipped data, or projection coefficient) and say what the bias-aware interval guarantees under heteroskedasticity, which is still open. 28
12 r-1: optionally set it to the exact (1 - alpha-L) quantile of lambda-max of the noise, computed offline by Monte Carlo (7.43 sigma-1 instead of 11.53 sigma-1 for p = 11), at no privacy cost. 28
13 The evaluation settings f-1 = 0.1, theta-star = 2.5 and kappa-Psi = 2 were chosen by simulation of earlier variants; confirm them under the two-stage rule. 29
14 Say whether the coverage statements below are conditional on passing the gate. . . 29
15 Consider centring the intervals at the bootstrap bias-corrected estimate 2 theta-tilde minus the bootstrap mean, as the proof review suggests. 30
16 The threshold treats lambda-min of the noisy stage-1 Gram as n times ell and does not model its downward bias under noise; either derive a correction (it depends on how many eigenvalues of Sigma lie near ell) or keep the empirical factor 1.2. 32
17 Decide whether the exact-quantile r-1 and f-1 near 0.35 become the recommended settings (they lower the threshold by 37 percent together); coverage would need to be re-checked. 33
18 Decide whether feature screening on public reference data becomes a Setup step, and what reference data it requires. 33
19 Decide whether this section stays in the main text, given that the correction is off in the evaluation, or moves to an appendix. 35
20 This section is carried over from the earlier draft and adapted to threshold keys. Split inference is not part of the v3.1 specification: the sensitivity, noise and ledger charge of a prediction release are still to be defined. 35
21 Rerun Sections 13.4 to 13.8 under v3.1: two releases with the tight conversion, the ridge from stage 1, delta = 1e-9 (or the chosen default), the bias correction off, and the gate. 39
22 The coverage table uses noise levels set directly (sigma-1 = 2.5 sigma-2). Repeat it with the calibrated sigma-1 and sigma-2 of a real budget, and on real data. 41
23 Coverage at epsilon = 8 (0.927) and for some protein settings (46 of 50) is slightly below 0.955. Repeat with about 100 runs per setting before drawing conclusions. . 42
24 Regenerate the figures with the current notation (theta-0 instead of rho-0) when they are rerun. 44
25 Real-data preprocessing: Wang's pipeline normalizes whole rows across both parties' features, which no party can do in the vertical setting, and scales y by the data maximum. Redo the comparison under v3.1 with per-party preprocessing (block normalization) and bounds from public reference data, against AdaSSP with Wang's joint normalization. 45

26 Reconcile AdaSSP's published noise constants with ours (possibly a looser adjacency in Wang's calibration constant); until then the published column may overstate AdaSSP's advantage. 45
27 Run the staged benchmarks at 100k, 300k and 1M rows once the PSI item length and the datasets are decided. 49
28 The proofs document and the v3.1 specification are not yet in the repository; add them (for example under docs/) so that this paper can cite them. 53

1 Introduction

Organizations such as Statistics Canada (StatCan) hold administrative and survey datasets that span the full Canadian population. University researchers, by contrast, often hold smaller cohort datasets enriched with domain-specific variables—clinical measurements, survey responses, or behavioural signals—that StatCan does not collect. Neither dataset alone is sufficient to answer certain research questions: a policy researcher studying labour-market outcomes might hold a cohort's health and education records, while StatCan holds their tax filing history, employment insurance claims, and income trajectories. The scientific value of joining these datasets is evident; the barrier to doing so is equally clear. Privacy law, institutional policy, and the practical realities of data governance make direct data sharing infeasible.

The Research Data Centre (RDC) model partially addresses this by letting researchers work with anonymized StatCan data inside a controlled environment. However, it does not permit a researcher to bring their own dataset and perform a record linkage against StatCan records in order to jointly train a statistical model—a process that typically requires six to eight weeks of vetting, human review, and institutional approvals. This overhead is a meaningful bottleneck on the pace and scope of Canadian policy research.

This document proposes a cryptographic protocol for *vertical federated learning* (VFL) between a researcher and StatCan. The term "vertical" refers to the structure of the data: both parties hold overlapping sets of individuals (rows) but non-overlapping sets of variables (columns). The goal is to jointly fit a regression model over the combined column space without either party ever observing the other's raw data. The two datasets are linked by a common identifier, but the linkage itself must remain private: StatCan should not learn which of its records belong to the researcher's cohort.

Scope. We treat the case in which **StatCan holds the response variable y** (e.g. tax-reported income, employment status, benefit receipt), and the researcher wishes to learn whether their cohort-specific covariates predict that outcome, using StatCan's covariates as controls. Both parties are honest-but-curious.

The design in one paragraph. Linkage is labeled private set intersection (PSI) on oblivious-PRF values of the identifiers, after which R knows its matches and their positions in O's secretly shuffled register. R computes its own Gram block exactly and uploads its features, aligned to O's order and zero elsewhere, encrypted under its share of a two-party CKKS key. O converts them to the joint key and multiplies them by its own plaintext columns, which gives encryptions of every statistic that involves its data. A release is a threshold decryption in which O computes its share of the decryption, adds discrete-Gaussian noise to it, and sends it without the part of the ciphertext that would let R remove the noise. O never decrypts anything, R never sees a noiseless statistic that depends on O's data, and every decryption needs O's cooperation, so O can refuse once the budget is spent.

Contributions.

1. **Protocol.** A two-party protocol that combines OPRF-based labeled PSI, a seeded upload under R's key share converted by O, plaintext–ciphertext aggregation, and a release in which O's decryption share carries the DP noise (Section 6, Algorithms 1–7).

2. **Privacy.** The exact replace-one sensitivity of the released statistics; a lattice discrete-Gaussian release that is exactly ρ_s-zCDP and statistically floods every encryption error term; simulation-based output privacy; input and cohort privacy (Sections 7 and 8).
3. **Statistics.** A two-stage ridge whose choice is independent of the noise in the estimation release; two named estimands; a Wald interval from a parametric bootstrap of the mechanism; a bias-aware interval for the regression coefficients; and the leading-order mechanism bias with an optional correction (Sections 9 and 10).
4. **Preprocessing.** Per-party transforms with public constants that make typical records fill the clipping bounds, which no joint computation is needed for (Section 6.1).
5. **Evaluation.** Exactness of the cryptographic layer, the calibration of the two-release protocol, interval coverage, end-to-end runs of a full implementation on real and synthetic data, coefficient accuracy against the intersection size, and a benchmark against AdaSSP on 29 UCI datasets (Section 13).

2 Related work

[**Check 2:** All bibliographic details in this section were written from memory and must be verified. The positioning against Froelicher et al. also needs a careful read of their papers.]

Vertical federated regression. Linear and logistic regression on vertically partitioned data has been built from additively homomorphic encryption with entity resolution [18] and from secure multiparty computation [17]; see [32] for a survey. These systems typically assume a coordinator or reveal the exact fitted model, and they do not give the output a differential-privacy guarantee.

Multiparty and threshold homomorphic encryption. Threshold fully homomorphic encryption goes back to Asharov et al. [3]. Mouchet et al. [24] give practical n-out-of-n multiparty schemes from RLWE, with collective key generation and collective key switching, which we use with two parties. Froelicher et al. [15, 16] build federated analytics on multiparty HE and add differential-privacy noise when results are collectively decrypted. Our release follows the same principle of attaching the noise to the decryption. Our setting differs: the data are vertically partitioned, records must first be linked privately, one party is the only analyst, and we give the exact sensitivity under the resulting adjacency, the flooding analysis for CKKS, and valid inference.

Differentially private linear regression. Sufficient-statistics perturbation is the standard approach [12, 29]; AdaSSP [30] adds an adaptive ridge chosen from a private estimate of the smallest eigenvalue. These methods assume one curator who holds the joined table. Our ridge rule borrows the idea of choosing λ from a private eigenvalue bound, but takes the bound from a separate release so that the estimation noise is independent of λ.

Private set intersection. Labeled PSI from homomorphic encryption [7, 8, 10] builds on polynomial evaluation [14] and cuckoo hashing [25]. We hash identifiers with the 2HashDH oblivious PRF [19], standardized in RFC 9497 [26]. Circuit-PSI [27] would hide the match set from R as well (Remark 15.1).

Homomorphic encryption and differential privacy. Li and Micciancio [20] showed that decrypted CKKS values can leak the secret key, and Li, Micciancio, Schultz and Sorrell [21] use differential privacy to protect approximate HE. Our release floods the error statistically with noise that is also the DP noise of the statistic (Section 7).

3 Setting and notation

O restricts its register to a study scope $D_\Phi = \{j : \Phi(j)\}$, where the predicate Φ uses public attributes only (for example province and birth-year band), and pads D_Φ with dummy rows—fresh random identifiers that never match, with $\mathbf{x}_O = \mathbf{0}$ and $y = 0$—to a public size $\bar{n}_O$. R holds $n_R \le n_R^{\max}$ cohort records. Let I be the cohort members that appear in D_Φ and $n = |I|$. Table 1 collects the notation.

Table 1: Notation. ρ always denotes a zCDP budget, Δ a sensitivity, Ω the CKKS scale and ϑ a validity ratio.

Symbol	Meaning
Φ, $\bar{n}_O$, $n_R^{\max}$	study scope (public attributes); padded register size; certified cohort cap
n, I, τ, π	matched records; R's matched cohort members; O's secret shuffle of its padded rows; alignment $\pi(i) = \tau(j)$ for $i \in I$
d_R, d_O, p	features at R and at O; $p = d_R + 1 + d_O$, the $+1$ being R's intercept
$\dot{\mathbf{x}}_i$	R's row with the intercept, $(1, \mathbf{x}_{R,i})$; $\dot{\mathbf{X}}_R$ stacks these rows in τ-order, zero off the match
$\mathbf{b} \in \{0,1\}^{\bar{n}_O}$	selection vector: $b_t = 1$ iff $t \in \pi(I)$
B_R, B_O, B_y	clipping bounds: $\|\dot{\mathbf{x}}\| \le B_R$ (intercept included), $\|\mathbf{x}_O\| \le B_O$, $\|y\| \le B_y$; $r, o, w = B_R^2, B_O^2, B_y^2$
Δ, $\Delta_{\neg yy}$, Δ_G, Δ_y	replace-one ℓ_2 sensitivities in the slot domain: full release with and without $\mathbf{y}^\top\mathbf{y}$; (triu $\mathbf{B}, \mathbf{C}$) only; $\mathbf{y}^\top\mathbf{y}$ alone ($= w$)
$(\varepsilon_{\text{run}}, \delta_{\text{run}})$, f_1, f_y	per-run budget; stage-1 share; share of a separate $\mathbf{y}^\top\mathbf{y}$ release
ρ_s, ρ_{run}	zCDP cost of release s; $\rho_{\text{run}} = \sum_s \rho_s$
γ, δ_{DP}, ζ	statistical slack of a run; the part of δ_{run} left to the DP mechanism; the largest share of δ_{run} the slack may take
$(\varepsilon_L, \delta_L)$, ζ_L; ρ_L, Γ_L	ledger cap of a base register and its slack share; the derived zCDP and slack capacities
N, $N_s = N/2$, q_0	ring dimension, slots, last-level modulus
Ω_U, Ω_u, Ω_V, Ω_f	encoding scales of R's vectors, of the one-hot masks and of O's data; the exact final scale $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1q_2)$
Ecd, Dcd	encoding $\mathsf{Ecd}_\Omega(\mathbf{z}) = \lfloor \Omega\, \tau_{\text{can}}^{-1}(\mathbf{z}) \rceil$ and decoding $\mathsf{Dcd}(v) = \tau_{\text{can}}(v)/\Omega_f$, with τ_{can} the canonical embedding
$\sigma_{c,s}$, σ_s	noise scale on the coefficient lattice; the induced slot-domain s.d. $\sigma_s = \sigma_{c,s}\sqrt{N/2}/\Omega_f$
$B_{\text{tot},s}$, δ_B, κ	ℓ_2 bound on all flooded error terms of release s, holding with probability $\ge 1 - \delta_B$; flooding parameter (statistical distance $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ per release), derived by Calibrate
ϑ, $\vartheta^\star$	validity ratio $\lambda_{\min}(\hat{\mathbf{G}} + \lambda\Psi)/(2\sigma_2\sqrt{p})$; its target
$\mathcal{E}$, $\mathcal{E}_G$, $\mathcal{E}_2$	released entries: all; (triu $\mathbf{B}, \mathbf{C}$); stage-2 entries ($\mathbf{y}^\top\mathbf{y}$ only in joint mode)

Parameters. The protocol, its algorithms and its theorems are stated for a parameter vector $\mathcal{P}$: security parameters (ring dimension, modulus chain, δ_B, the floor $\kappa_{\min}$ on the flooding parameter, the security level λ_{sec}), privacy parameters (the budget, the splits f_1 and f_y, the slack share ζ, the ledger cap), statistical parameters, the data bounds and preprocessing, the PSI parameters, and the encoding scales Ω_U and Ω_u. Section 12 lists every parameter with its role, the constraint it must satisfy and the value used in our evaluation, which we call the *evaluation setting*; the evaluation settings are not part of the protocol. CALIBRATE derives the flooding parameter κ, the noise, the scales and the size of q_0 from $\mathcal{P}$, or refuses the run (Section 7.3). Numbers in Sections 6–10 are evaluation settings or worked examples and are labelled as such.

The target is a ridge fit on the matched, column-joined data $\mathbf{X}^I = [\dot{\mathbf{X}}_R^I \mid \mathbf{X}_O^I]$ with

$$\mathbf{G} := \mathbf{X}^{I\top}\mathbf{X}^I = \begin{bmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{B} \end{bmatrix}, \qquad \mathbf{c} := \mathbf{X}^{I\top}\mathbf{y}^I = \begin{bmatrix} \mathbf{c}_R \\ \mathbf{c}_O \end{bmatrix}, \tag{1}$$

where $\mathbf{A}$ is R's own block and $\mathbf{B}$, $\mathbf{C}$, $\mathbf{c}_R$, $\mathbf{c}_O$ and $\mathbf{y}^\top\mathbf{y}$ depend on O's data. We write Π_R and Π_O for the orthogonal projectors onto the two coordinate blocks, so $\Pi_R + \Pi_O = \mathbf{I}_p$. The two estimands, the ridge target $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$ and the regression coefficient $\boldsymbol{\beta}$, are defined in Section 9.1.

4 Threat model and guarantees

Both parties are *honest-but-curious*. They follow the protocol, including key generation and R's preparation of its inputs (clipping, the selection vector), and they try to learn what they can from their views. Channels are private and authenticated. R is the analyst and receives the output.

Definition 4.1 (Datasets and adjacency)**.** A dataset is O's padded scope register $D = \{(\text{id}_j, \mathbf{x}_{O,j}, y_j)\}_{j \le \bar{n}_O}$, with the cohort $(\text{id}, \mathbf{X}_R)$, the identifiers and Φ held fixed. $D \sim D'$ if they differ in $(\mathbf{x}_O, y)$ of exactly one row, matched or not (*replace-one* adjacency).

Definition 4.2 (Zero-concentrated DP)**.** A randomized mechanism $\mathcal{M}$ is ρ-zCDP if the Rényi divergence of order a between $\mathcal{M}(D)$ and $\mathcal{M}(D')$ is at most ρa for all $a > 1$ and all $D \sim D'$ [5]. It is (ε, δ)-DP if $\Pr[\mathcal{M}(D) \in S] \le e^\varepsilon \Pr[\mathcal{M}(D') \in S] + \delta$ for all $D \sim D'$ and measurable S.

Definition 4.3 (SIM-CDP and IND-CDP, after [22])**.** A protocol is (ε, δ)-*SIM-CDP for* R *with statistical slack* γ if there exist a mechanism $\mathcal{M}$ that is (ε, δ)-DP with respect to $\sim$ and a probabilistic polynomial-time simulator $\mathcal{S}$ such that, for every D and every PPT distinguisher $\mathcal{A}$,

$$\left|\Pr[\mathcal{A}(\text{view}_\mathsf{R}(D)) = 1] - \Pr[\mathcal{A}(\mathcal{S}(\mathcal{M}(D))) = 1]\right| \le \gamma + \text{negl}(\lambda_{\text{sec}}).$$

It is (ε, δ)-*IND-CDP* if $\Pr[\mathcal{A}(\text{view}_\mathsf{R}(D)) = 1] \le e^\varepsilon \Pr[\mathcal{A}(\text{view}_\mathsf{R}(D')) = 1] + \delta + \text{negl}(\lambda_{\text{sec}})$ for all $D \sim D'$ and every PPT $\mathcal{A}$.

Mironov et al. state these notions for pure ε; the extension to (ε, δ) is the standard one, and the slack γ makes explicit the statistical terms that are not negligible in λ_{sec} at a concrete flooding parameter.

Lemma 4.1 (From simulation to indistinguishability)**.** (ε, δ)-*SIM-CDP with slack* γ *implies* $(\varepsilon, \delta + (1 + e^\varepsilon)\gamma)$-*IND-CDP.*

Proof. Chain $\text{view}(D) \approx \mathcal{S}(\mathcal{M}(D))$ (cost γ), the DP of $\mathcal{S} \circ \mathcal{M}$ (post-processing; factor e^{ε}, cost δ), and $\mathcal{S}(\mathcal{M}(D')) \approx \text{view}(D')$ (cost $e^{\varepsilon}\gamma$). □

Table 2: Guarantees of v3.1 against honest-but-curious parties. Correctness assumes honest computation.

	Guarantee	Basis
G1	**R-input privacy**: O's view can be simulated from $(n_R^{\max}, \bar{n}_O)$ and the public parameters.	RLWE in s_R; security of the PSI. O never receives a decryption share from R.
G2	**Cohort privacy**: O learns nothing about R's cohort beyond $n_R^{\max}$.	Padded OPRF queries; labeled PSI.
G3	**Output privacy**: R's view of a run is $(\varepsilon_{\text{run}}, \delta_{\text{DP}})$-SIM-CDP for $\sim$ with slack γ, hence $(\varepsilon_{\text{run}}, \delta_{\text{run}})$-IND-CDP. The joint view of all researchers over all runs on a base register is $(\varepsilon_L, \delta_L)$-IND-CDP.	Theorem 7.2; Proposition 7.2 (ledger).
G4	**Disclosed**: R learns I (which cohort members are in D_Φ) and uniformly random positions $\tau(j)$, and nothing else about O's register.	By design (Remark 4.1); OPRF-keyed items and labels and SanitizeBFV (Section 8).

Remark 4.1 (What is revealed by design)**.** R learns which of its cohort appear in O's scope, and n. When membership is itself sensitive (e.g. a benefits register, where membership *is* the outcome), this must be disclosed to data stewards. DP does not protect membership or n, nor R's own cohort in a published model, since $\mathbf{A}$ is exact. A membership-private variant is sketched in Remark 15.1.

5 Building blocks

5.1 Two-party threshold CKKS

CKKS [9] encrypts vectors of $N/2$ real numbers. A ciphertext is a pair (c_0, c_1) of polynomials modulo q, and decryption with the secret key s gives

$$c_0 + c_1 s = \mathsf{Ecd}(\mathbf{z}) + e \pmod{q},$$

the encoded vector plus a small error. In the threshold variant the key is split additively, $s = s_R + s_O$, so decryption splits as $(c_0 + c_1 s_O) + c_1 s_R$: the first term needs O and the second needs R.

Key generation. Following Mouchet et al. [24], from a common random polynomial a each party draws its share and an error and publishes one polynomial,

$$p_{0,R} = -a s_R + e_R, \qquad p_{0,O} = -a s_O + e_O, \qquad pk = \big(p_{0,R} + p_{0,O},\ a\big) = (-as + e,\ a).$$

Rotation keys for the slot shifts 2^i are generated the same way. The circuit has depth two and no ciphertext–ciphertext products, so no relinearization or conjugation key is needed.

Upload at half size. R encrypts with its own share symmetrically, using a random a expanded from a short seed: $c_0 = -a s_R + e + \mathsf{Ecd}(\mathbf{v})$, $a = \text{PRG}(\varsigma)$. It sends (c_0, ς). O regenerates a and sets $c_0 \leftarrow c_0 - a s_O$, after which $c_0 + as = \mathsf{Ecd}(\mathbf{v}) + e$: a ciphertext under the joint key.

What O computes. O adds ciphertexts, multiplies them by its own plaintext columns, and rotates slots to sum a vector. None of this needs a key share.

5.2 Oblivious PRF

R obtains $F_k(x) = H_2(x, H_1(x)^k)$ under O's key k without O seeing x and without R learning k (2HashDH [19, 26]). R sends $\alpha = H_1(x)^r$ for a random r; O returns $\beta = \alpha^k$; R unblinds $\beta^{1/r} = H_1(x)^k$ and hashes. α is uniform in the group, so O learns nothing about x. The output is split into an item part $F_k(x)$, used for matching, and a key part $F'_k(x)$, used to open labels. The OPRF makes every membership test an online interaction with O, which matters because identifiers such as social insurance numbers are a small, guessable space.

5.3 Labeled PSI

In labeled PSI [7, 8, 10] the receiver learns which of its items are in the sender's set and, for each match, a label attached by the sender. O hashes its items into bins, interpolates per bin a membership polynomial whose roots are the items and a label polynomial that maps each item to its encrypted label, and evaluates both on R's encrypted, cuckoo-hashed items. Here the label is the row's position $\tau(j)$ in O's shuffled register. Appendix B gives the details.

5.4 zCDP and the discrete Gaussian

A mechanism that adds Gaussian noise of s.d. σ to a statistic with ℓ_2 sensitivity Δ is exactly $\Delta^2/(2\sigma^2)$-zCDP, and so is the multivariate discrete Gaussian on $\mathbb{Z}^N$ [5, 6]. zCDP costs add under composition. A ρ-zCDP mechanism is (ε, δ)-DP with $\varepsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$; the tight conversion of Canonne, Kamath and Steinke [6] allows a larger ρ at the same (ε, δ) (about 27% larger at $\varepsilon = 1$, $\delta = 10^{-9}$). Floating-point samplers leak through their low-order bits [23]; we use an exact integer sampler.

6 The protocol

Figure 1 shows the message flow and Algorithms 1–7 give the pseudocode.

Phase 0: setup (Algorithm 1). The parties agree on the parameter vector $\mathcal{P}$ (Section 12): the scope, the caps, the budget, the ledger cap of the base register, the statistical constants, the PSI parameters and the encoding scales. Each party preprocesses its own columns with public constants (Section 6.1), and the clipping bounds are chosen on the preprocessed public reference data or from domain knowledge. The sensitivities follow from the bounds alone (Section 7.1). O runs Calibrate (Algorithm 7), which derives the flooding parameter, the CKKS scale and the noise from $\mathcal{P}$ and refuses the run if $\mathcal{P}$ is not admissible; O also refuses if its ledger would exceed the cap (Section 7.4). R repeats Calibrate, whose outputs depend on public values only, and checks O's. The parties run threshold key generation, and O draws a fresh OPRF key, a fresh shuffle τ, and prepares its PSI polynomials.

[**Check 3:** ChooseBounds: the evaluation uses the 99.5% quantile of each block's norm on the preprocessed reference data. The optimized choice (quantile grid minimizing first-order MSE plus squared clipping bias) has not been specified or evaluated.]

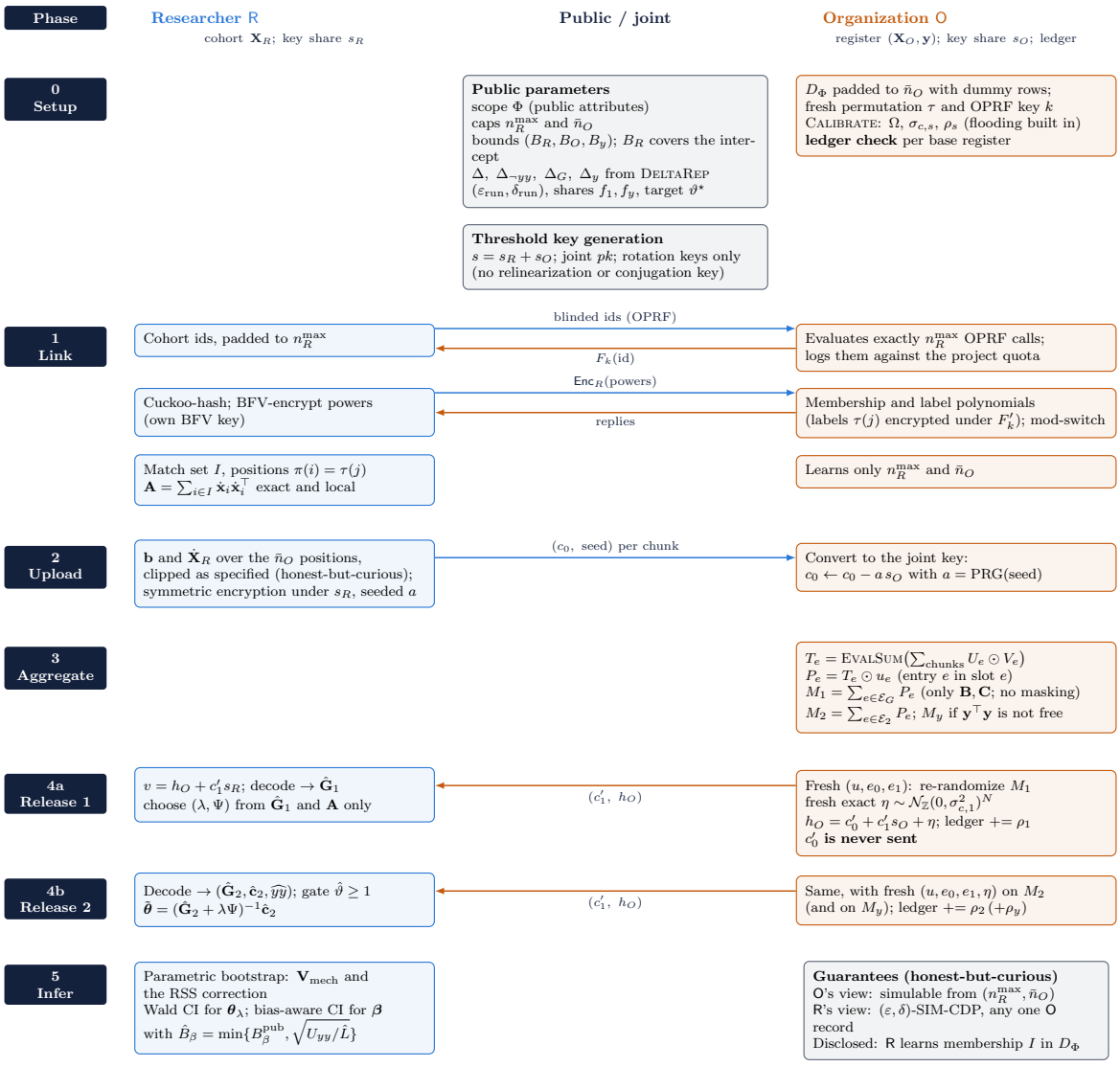


Figure 1: Message flow of protocol v3.1. Blue arrows go from R to O and orange arrows from O to R. O never receives a decryption share, and R never receives c'_0.

Phase 1: link (Algorithm 2). R pads its identifiers to $n_R^{\max}$, obtains their OPRF values, and runs labeled PSI. It learns I and the positions $\pi(i) = \tau(j)$, and computes its own block

$$\mathbf{A} = \sum_{i \in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$$

exactly and locally. $\mathbf{A}$ is never transmitted and never noised: it is a function of $(I, \mathbf{X}_R)$, both known to R, and it does not change between adjacent datasets.

Phase 2: upload (Algorithm 3). R writes the selection vector $\mathbf{b}$ and its features $\dot{\mathbf{X}}_R$ over all $\bar{n}_O$ positions, in τ-order and zero off the match, encrypts them under s_R with seeded randomness, and sends them in chunks of at most $N/2$ slots. O converts each ciphertext to the joint key.

Phase 3: aggregate (Algorithm 4). For each released entry O multiplies one of R's encrypted vectors by one of its plaintext columns and sums:

$$\mathsf{Enc}(B_{jk}) = \textstyle\sum_t \mathsf{Enc}(b_t)\ X_{O,tj} X_{O,tk}, \qquad \mathsf{Enc}(C_{jk}) = \textstyle\sum_t \mathsf{Enc}\Big(\dot{X}_{R,tj}\Big)\ X_{O,tk}, \tag{2}$$

$$\mathsf{Enc}(c_{O,k}) = \textstyle\sum_t \mathsf{Enc}(b_t)\ X_{O,tk}\, y_t, \qquad \mathsf{Enc}(c_{R,j}) = \textstyle\sum_t \mathsf{Enc}\Big(\dot{X}_{R,tj}\Big)\ y_t, \tag{3}$$

and $\mathsf{Enc}\big(\mathbf{y}^\top \mathbf{y}\big) = \sum_t \mathsf{Enc}(b_t)\ y_t^2$. Unmatched positions contribute zero to the values, because $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ are zero there. O then packs the results so that each released entry sits alone in its own slot: M_1 holds only triu $\mathbf{B}$ and $\mathbf{C}$, for stage 1, and M_2 holds every stage-2 entry. A separate M_y holds $\mathbf{y}^\top \mathbf{y}$ when it is not free (Section 7.1).

Phase 4: release (Algorithm 5). For each packed ciphertext $M = (c_0, c_1)$, O re-randomizes it under the joint public key with fresh (u, e_0, e_1), computes its noisy decryption share

$$h_O = c_0' + c_1' s_O + \eta, \qquad \eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_c^2)^N \text{ fresh},$$

charges its ledger, and sends (c_1', h_O). It never sends c_0'. R computes $v = h_O + c_1' s_R = \mathsf{Ecd}(\mathbf{z}) + e + \eta$ and decodes. Stage 1 releases M_1 with s.d. σ_1, stage 2 releases M_2 with s.d. σ_2, and the releases use independent randomness.

Phase 5: solve and infer (Algorithm 6). From stage 1 and its exact $\mathbf{A}$, R chooses the ridge (λ, Ψ). After stage 2 it checks the gate $\hat{\vartheta} \geq 1$, solves $\tilde{\boldsymbol{\theta}} = (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1} \hat{\mathbf{c}}_2$, and computes both intervals (Section 9). All of this is post-processing.

Remark 6.1 (Changes relative to earlier drafts)**.** (i) **Keys.** R no longer holds the CKKS secret key. The key is shared, O adds the noise inside its decryption share, and c_0' is never transmitted, which closes a leak of O's key share.

(ii) **Flooding.** The DP noise η itself floods every error term that depends on O's data or secrets, statistically and in ℓ_2. The earlier separately calibrated "error channel" and its cost ρ_{err} are gone.

(iii) **Two releases.** The ridge is chosen from a first release instead of from a committed public eigenvalue bound, so it adapts to the data while staying independent of the estimation noise.

Algorithm 1 SETUP (Phase 0)

Both: agree on the parameter vector $\mathcal{P}$ (Table 9): the scope Φ (public attributes only), the caps $n_R^{\max}$ and $\bar{n}_O$, the budget $(\varepsilon_{\text{run}}, \delta_{\text{run}})$, the ledger cap $(\varepsilon_L, \delta_L, \zeta_L)$, f_1 (and f_y), ζ, $\kappa_{\min}$ (or a fixed κ), δ_B, t_η, $\vartheta^\star$, $\alpha_{\text{CI}}, \alpha_L, \alpha_U$, B_β^{pub}, κ_Ψ, η_B, B, the PSI parameters, N, the chain, Ω_U and Ω_u

O: $D_\Phi \leftarrow \{j : \Phi(j)\}$; pad with dummy rows (fresh random identifiers, $\mathbf{x}_O = \mathbf{0}$, $y = 0$) to $\bar{n}_O$; draw a fresh secret permutation τ of $[\bar{n}_O]$

R: and **O:** , each on its own columns only: PREPROCESS with constants from public reference data or domain knowledge

Both: $(B_R, B_O, B_y) \leftarrow$ CHOOSEBOUNDS on the preprocessed reference data, or from domain knowledge ▷ B_R covers R's intercept

$(\Delta, \Delta_{\neg yy}) \leftarrow \text{DELTAREP}(B_R, B_O, B_y)$; $\Delta_G \leftarrow \text{DELTAREP}(B_R, B_O, 0)$; $\Delta_y \leftarrow B_y^2$

mode $\leftarrow$ joint if $\Delta - \Delta_{\neg yy} \le 10^{-9}\Delta$, otherwise separate ▷ release $\mathbf{y}^\top\mathbf{y}$ jointly only when it is free

O: $(\kappa, \Omega_V, \Omega_f, \{\sigma_{c,s}, \rho_s, \gamma_s\}_s) \leftarrow \text{CALIBRATE}(\mathcal{P})$; refuse if it refuses ▷ R repeats it and checks

O: **if** the ledger of the base register does not admit $(\rho_{\text{run}}, \gamma)$ (Section 7.4) **then** refuse

Both: threshold key generation with a common random a: $s = s_R + s_O$, $pk = (p_0, a)$ with $p_0 = -as + e$, and rotation keys for $\{2^i\}$
(no relinearization or conjugation key)

O: fresh OPRF key k; PSI preprocessing on the items $\{F_k(\text{id}_j)\}$ with labels $\tau(j)$ encrypted under keys derived from $F'_k(\text{id}_j)$; bin sizes and partition counts public; log the run against the OPRF quota of (researcher, project)

Algorithm 2 LINK: labeled PSI with an OPRF (Phase 1)

R: pad the cohort identifiers with random dummies to exactly $n_R^{\max}$

R: run the OPRF with O on each padded identifier $\to F_k(\text{id}), F'_k(\text{id})$ ▷ O evaluates exactly $n_R^{\max}$ calls

R: cuckoo-hash the items $F_k(\text{id})$ into the bins; send BFV encryptions of their windowed powers under R's own BFV key

O: evaluate the membership and label polynomials of every bin and partition; mask; SANITIZEBFV; reply ▷ flooding: κ_ψ

R: decrypt $\to$ the match set I and positions $\pi(i) = \tau(j)$ for $i \in I$ (labels opened with keys from $F'_k(\text{id}_i)$)

R: $\mathbf{A} \leftarrow \sum_{i \in I} \dot{\mathbf{x}}_i \dot{\mathbf{x}}_i^\top$, exact and local ▷ never transmitted, never noised

Algorithm 3 UPLOAD (Phase 2)

for $t = 1, \ldots, \bar{n}_O$ **do**

  R: $b_t \leftarrow \mathbb{1}[t \in \pi(I)]$; $\dot{\mathbf{x}}_t \leftarrow \big(1,\ \text{clip}(\mathbf{x}_{R,i}; \sqrt{B_R^2 - 1})\big)$ if $t = \pi(i)$, otherwise $\mathbf{0}$

end for

$m \leftarrow \lceil \bar{n}_O / N_s \rceil$ ▷ chunks of at most $N_s = N/2$ slots

for each vector $\mathbf{v} \in \{\mathbf{b}, \dot{\mathbf{x}}_{\cdot 0}, \ldots, \dot{\mathbf{x}}_{\cdot d_R}\}$ and each chunk $h \le m$ **do**

  R: draw a seed ς; $a \leftarrow \text{PRG}(\varsigma)$; $c_0 \leftarrow -a\, s_R + e + \text{Ecd}_{\Omega_U}(\mathbf{v}[h])$, with e from the RLWE error distribution; send (c_0, ς)

  O: $a \leftarrow \text{PRG}(\varsigma)$; $c_0 \leftarrow c_0 - a\, s_O$ ▷ (c_0, a) now encrypts $\mathbf{v}[h]$ under $s = s_R + s_O$

end for

R encodes at the parameter Ω_U (Section 7.3)

Table 3: What crosses between the parties. All messages travel over a mutually authenticated channel with structured (non-executable) serialization.

$\mathsf{R} \to \mathsf{O}$	key-generation shares ($p_{0,R}$ and rotation-key shares); $n_R^{\max}$ blinded OPRF elements; BFV encryptions of windowed powers under R's own BFV key; seeded uploads (c_0, ς) of $\mathbf{b}$ and the columns of $\dot{\mathbf{X}}_R$
$\mathsf{O} \to \mathsf{R}$	public parameters: the agreed $\mathcal{P}$ (including $B_O, B_y, d_O, \bar{n}_O$) and CALIBRATE's outputs $(\kappa, \Omega_V, \Omega_f, q_0, \sigma_{c,s})$, PSI sizes; key-generation shares; OPRF responses; masked, sanitized PSI replies; per release, (c_1', h_O)
never transmitted	s_R, s_O; $\mathbf{A}$; raw $\mathbf{X}_R$, $\mathbf{X}_O$, $\mathbf{y}$; τ; k; c_0', u, e_0, e_1 and η of every release

(iv) **Accounting.** zCDP with the tight conversion replaces the analytic Gaussian calibration of a single release.

(v) **Linkage.** An OPRF with a fresh key per run replaces a keyed hash with a key known to R, and OPRF queries are padded and counted against a quota.

(vi) **Input checks.** The homomorphic input checks of v3 are removed: they addressed only a malicious researcher, and one could be bypassed. Robustness to a deviating researcher is an extension (Section 15).

(vii) **Bias correction.** The zero-budget correction of Section 10 is now an option, off in our evaluation.

(viii) **Preprocessing.** Each party transforms its own columns in Setup, with public constants (Section 6.1).

(ix) **Parameters.** The protocol is stated for a parameter vector with explicit admissibility constraints (Section 12). The flooding parameter is derived from the budget and the ledger cap, the PSI replies are sanitized with their own flooding parameter, and the ledger has a specified composition rule.

6.1 Per-party preprocessing

The noise is sized to the clipping bounds, so every record whose norm is well below its bound wastes part of the budget. After standardization, real features are often heavy-tailed. On the protein dataset of Section 13.4, the 99.5% quantile of the squared norm of R's block (with the intercept) is 9.5 times its median, and that of O's block 19 times. Bounds at these quantiles leave typical records using a small part of the budget. Setup therefore includes a preprocessing step, PREPROCESS in Algorithm 7, in which each party transforms its own columns:

(i) standardize each column with public constants, from public reference data or domain knowledge;

(ii) optionally, a per-column tail transform with public constants: winsorizing at $|z| \le c$, a rank-Gaussian transform from public reference quantiles, or a log transform of a skewed variable;

(iii) optionally, *block normalization*: each party scales its own block of every row to unit norm (R excludes the intercept, so $\|\dot{\mathbf{x}}\|^2 = 2$; O gets $\|\mathbf{x}_O\| = 1$). A clipped log of the block norm may be appended as an extra feature to keep the magnitude.

O standardizes y, and may winsorize it, in the same way. The bounds are then chosen on the transformed reference data. For block normalization they are exact: $B_R^2 = 2$, $B_O = 1$.

Proposition 6.1 (Preprocessing preserves the guarantees)**.** *Let each party's transform be a fixed function of its own record and of public constants. Then the protocol run on the transformed data, with bounds that hold on the transformed domain, has the guarantees of Table 2 with respect to replace-one adjacency on the raw data.*

Proof. O's transform maps one raw record to one transformed record, so changing one raw record $(\mathbf{x}_O, y)$ changes exactly one transformed record, which still satisfies the bounds; R's records are fixed under $\sim$. The sensitivity of Section 7.1 therefore bounds the change, and the rest of the analysis is unchanged. The constants are data-independent, as CALIBRATE's outputs must be. R may standardize with its own constants, since its data are not protected against itself, but it commits to B_R. □

The estimand is the regression on the transformed features. Winsorizing and log transforms keep the coefficients interpretable. Block normalization changes their meaning: each block contributes through its direction only, unless the log-norm feature is appended. It is therefore the choice for prediction rather than for interpreting coefficients.

Why not joint row normalization. Wang's preprocessing for AdaSSP normalizes each whole row $(\mathbf{x}_R, \mathbf{x}_O)$, which needs both parties' columns. We considered two ways to do this here.

- *Exact joint clipping* computes the scale factor $\min\{1, B/\|\mathbf{x}\|\}$ under encryption from R's encrypted row norms and O's plaintext ones. This needs a high-degree polynomial approximation and ciphertext–ciphertext products. Worse, scaling R's part by a factor that depends on O's data makes $\mathbf{A}$ depend on O's data, so $\mathbf{A}$ could no longer be exact.
- *Clipping only O's part to the remaining budget* $B^2 - \|\dot{\mathbf{x}}\|^2$ keeps $\mathbf{A}$ exact and the circuit linear. R assigns each matched record to one of K norm classes and uploads its selection vector and features split by class, still encrypted, and O pre-clips its columns once per class in plaintext, at about K times the upload and aggregation cost. On protein it does not pay. A joint budget alone raises Δ from 87.3 to 102, because the sensitivity is dominated by O's own norm, which enters twice, and records with a small R-norm may then use more of it. Keeping O's separate cap as well lowers Δ only to 82, about 12% fewer records.

The gap is caused by heavy tails under loose bounds, not by the lack of a joint bound, and per-party transforms close most of it. In end-to-end runs on protein, block normalization brings private test error within 2–6% of non-private, against 12–21% with standardization alone (Table 15).

[**Check 6:** Preprocessing defaults: which transform to recommend for prediction and for interpretation; the appended log-norm feature and winsorizing y are not yet evaluated.]

Algorithm 4 AGGREGATE and pack (O, Phase 3)

Input: entry map $e \mapsto (U_e, V_e)$, with R's ciphertext vector U_e and O's plaintext vector V_e in τ-order:
$\mathbf{B}_{jk}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,j} \odot \mathbf{x}_{O,k})$; $\mathbf{C}_{jk}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{x}_{O,k})$; $\mathbf{c}_{R,j}$: $(\mathsf{Enc}(\dot{\mathbf{x}}_j),\ \mathbf{y})$;
$\mathbf{c}_{O,k}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{x}_{O,k} \odot \mathbf{y})$; $\mathbf{y}^\top\mathbf{y}$: $(\mathsf{Enc}(\mathbf{b}),\ \mathbf{y} \odot \mathbf{y})$

1: **for** each $e \in \mathcal{E}$ **do**
2: $T_e \leftarrow \textsc{EvalSum}\big(\textsc{Rescale}\big(\sum_{h \le m} U_e[h] \odot \mathsf{Ecd}_{\Omega_V}(V_e[h])\big)\big)$ ▷ sum the chunks first; one EVALSUM per entry
3: $P_e \leftarrow \textsc{Rescale}\big(T_e \odot \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_{\mathrm{slot}(e)})\big)$ ▷ $\mathbf{u}_k$: one-hot slot k; entry e sits alone in its slot
4: **end for**
5: $M_1 \leftarrow \sum_{e \in \mathcal{E}_G} P_e$; $M_2 \leftarrow \sum_{e \in \mathcal{E}_2} P_e$; $M_y \leftarrow P_{\mathbf{y}^\top\mathbf{y}}$ if $mode = \mathsf{separate}$ ▷ depth 2; M_1 depends only on triu $\mathbf{B}$ and $\mathbf{C}$
Scales: R's vectors at Ω_U, the masks at Ω_u (parameters), O's data at Ω_V (from CALIBRATE); the final scale $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1q_2)$ is tracked exactly, and R decodes with it

Algorithm 5 RELEASE$(M, \sigma_c, \rho, \gamma_s)$: one DP release (Phase 4)

1: **O:** sample fresh u from the key distribution and e_0, e_1 from the error distribution
2: **O:** $c_0' \leftarrow c_0 + p_0u + e_0$; $c_1' \leftarrow c_1 + au + e_1$ ▷ re-randomize $M = (c_0, c_1)$ under the joint pk
3: **O:** $\eta \leftarrow (\textsc{SampleDG}(\sigma_c))^N$, fresh ▷ exact discrete Gaussian on $\mathbb{Z}^N$
4: **O:** $h_O \leftarrow c_0' + c_1's_O + \eta \bmod q_0$
5: **O:** Ledger += (ρ, γ_s); send (c_1', h_O); erase c_0', u, e_0, e_1, η ▷ c_0' is never sent
6: **R:** $v \leftarrow h_O + c_1's_R \bmod q_0$; **return** $\mathsf{Dcd}(v)$ restricted to the slots of this release ▷ decode with the exact final scale

Algorithm 6 SOLVE and INFER (R, Phase 5)

Input: $\mathbf{A}$, n, p, slot-domain s.d. σ_1, σ_2 (and σ_y); from $\mathcal{P}$: $\vartheta^\star$, $\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$, B_β^{pub}, κ_Ψ, bootstrap truncation η_B and size B

$(\hat{\mathbf{B}}_1, \hat{\mathbf{C}}_1) \leftarrow \textsc{Release}(M_1, \sigma_{c,1}, \rho_1, \gamma_1)$; $\hat{\mathbf{G}}_1 \leftarrow \begin{pmatrix} \mathbf{A} & \hat{\mathbf{C}}_1 \\ \hat{\mathbf{C}}_1^\top & \hat{\mathbf{B}}_1 \end{pmatrix}$

$r_1 \leftarrow \sigma_1\big(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)}\big)$ ▷ $\Pr[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1] \le \alpha_L$ (one-sided)

$\hat{L} \leftarrow \max\{0,\ \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1\}$; $g^\star \leftarrow 2\vartheta^\star \sigma_2 \sqrt{p}$

$\lambda_I \leftarrow \max\{0,\ g^\star + r_1 - \lambda_{\min}(\hat{\mathbf{G}}_1)\}$; $\lambda_O \leftarrow \min\{\lambda \ge 0 : \lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda \Pi_O) \ge g^\star + r_1\}$ (∞ if none)

[**Check 4:** Replace the bisection for lambda-O by the closed form of the proofs (Lemma S3(c)).]

$(\lambda, \Psi) \leftarrow (\lambda_O, \Pi_O)$ if $\lambda_O \le \kappa_\Psi \lambda_I$, otherwise $(\lambda_I, \mathbf{I})$ ▷ uses stage 1 and $\mathbf{A}$ only

$(\hat{\mathbf{B}}_2, \hat{\mathbf{C}}_2, \hat{\mathbf{c}}_R, \hat{\mathbf{c}}_O, \widehat{yy}) \leftarrow \textsc{Release}(M_2, \sigma_{c,2}, \rho_2, \gamma_2)$; if separate, $\widehat{yy} \leftarrow \textsc{Release}(M_y, \sigma_{c,y}, \rho_y, \gamma_y)$

assemble $\hat{\mathbf{G}}_2$, $\hat{\mathbf{c}}_2$; $\hat{\vartheta} \leftarrow \lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)/(2\sigma_2\sqrt{p})$; **if** $\hat{\vartheta} < 1$ **then return** refused ▷ budget is spent

$\tilde{\mathbf{P}} \leftarrow (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1}$; $\tilde{\boldsymbol{\theta}} \leftarrow \tilde{\mathbf{P}}\hat{\mathbf{c}}_2$ ▷ the bias correction of Section 10 is an option

$\widetilde{\mathrm{RSS}} \leftarrow \widehat{yy} - \hat{\mathbf{c}}_2^\top \tilde{\boldsymbol{\theta}} - \lambda\, \tilde{\boldsymbol{\theta}}^\top \Psi \tilde{\boldsymbol{\theta}}$; $\sigma_{yy} \leftarrow \sigma_2$ (joint) or σ_y (separate)

for $b = 1, \dots, B$ **do** ▷ parametric bootstrap of the mechanism: post-processing

  draw $\mathbf{E}^*$ (symmetric, s.d. σ_2 on the O-dependent entries, zero on $\mathbf{A}$), $\mathbf{f}^* \sim \mathcal{N}(0, \sigma_2^2 \mathbf{I})$, $g^* \sim \mathcal{N}(0, \sigma_{yy}^2)$

  keep the draw only if $\|\tilde{\mathbf{P}}^{1/2} \mathbf{E}^* \tilde{\mathbf{P}}^{1/2}\|_{\mathrm{op}} \le \eta_B$ ▷ truncation: otherwise the moments do not exist

  $\boldsymbol{\theta}_b^* \leftarrow (\hat{\mathbf{G}}_2 + \mathbf{E}^* + \lambda\Psi)^{-1}(\hat{\mathbf{c}}_2 + \mathbf{f}^*)$; $\mathrm{RSS}_b^* \leftarrow$ the RSS formula at $(\hat{\mathbf{G}}_2 + \mathbf{E}^*, \hat{\mathbf{c}}_2 + \mathbf{f}^*, \widehat{yy} + g^*)$

end for

$\mathbf{V}_{\mathrm{mech}} \leftarrow \widehat{\mathrm{Cov}}_b(\boldsymbol{\theta}_b^*)$; $\mathrm{Corr} \leftarrow \overline{\mathrm{RSS}^*} - \widetilde{\mathrm{RSS}}$; $\tilde{s}^2 \leftarrow \max\{\widetilde{\mathrm{RSS}} - \mathrm{Corr}, 0\}/(n-p)$

CI for the protocol estimand $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$: $\tilde{\theta}_j \pm z_{1-\alpha_{\mathrm{CI}}/2}\sqrt{(\mathbf{V}_{\mathrm{mech}})_{jj}}$

$\mathbf{V} \leftarrow \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2\, \tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$ ▷ adds sampling variance under the homoskedastic working model

$U_{yy} \leftarrow \min\big\{\widehat{yy} + z_{1-\alpha_U/2}\,\sigma_{yy} + B_y^2\sqrt{n\ln(2/\alpha_U)/2},\ nB_y^2\big\}$; $\hat{B}_\beta \leftarrow \min\{B_\beta^{\mathrm{pub}}, \sqrt{U_{yy}/\hat{L}}\}$ ▷ ∞ if $\hat{L} = 0$

[**Check 5:** U-yy as corrected by the proof review (the specification spent alpha-U twice and left sigma-yy undefined). Confirm.]

$\bar{b}_j \leftarrow \lambda\, \|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|_2\, \hat{B}_\beta$ ▷ worst-case ridge bias of coordinate j (first order)

CI for β_j (bias-aware): $\tilde{\theta}_j \pm \mathrm{cv}_{\alpha_{\mathrm{CI}}}\big(\bar{b}_j/\sqrt{V_{jj}}\big)\sqrt{V_{jj}}$, where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$ quantile of $|Z + t|$, $Z \sim \mathcal{N}(0,1)$

return $\tilde{\boldsymbol{\theta}}$, both intervals, (λ, Ψ), the shrinkage $-\lambda\tilde{\mathbf{P}}\Psi\tilde{\boldsymbol{\theta}}$, $\hat{\vartheta}$, $\sigma_2\,\mathrm{tr}(\tilde{\mathbf{P}})$, and the (ε, δ) spent

Algorithm 7 Subroutines

function DELTAREP(B_R, B_O, B_y) ▷ replace-one of $(\mathbf{x}_O, y)$, $\dot{\mathbf{x}}$ fixed; $r, o, w = B_R^2, B_O^2, B_y^2$; Proposition 7.1
  if $|r - w| \le 2o$: $\Delta^2 = (2o + r + w)^2/2 + 2rw$; **else if** $r - w \ge 2o$: $\Delta^2 = 4r(o + w)$
  else: exact candidate enumeration (three cubics), rounded up as a safeguard. Return Δ, and $\Delta_{\neg yy}$ from the same maximization without the $(y^2 - y'^2)^2$ term
  ▷ exact suprema for $d_O \ge 2$, valid but conservative for $d_O = 1$; $\Delta_G = (2o + r)/\sqrt{2}$ if $r \le 2o$, otherwise $2\sqrt{ro}$
end function

function CALIBRATE($\mathcal{P}$) ▷ public values only; refuses when a step has no solution
  mode, $n_{\text{rel}} \in \{2, 3\}$ and $\{\Delta_s\}$ as in SETUP; **refuse if** $\log_2(QP) > L_{\max}(N, \lambda_{\text{sec}})$
  $\gamma(k) := n_{\text{rel}}\big(\delta_B + 2^{-k}(1 + 2^{-80}) + 2^{-97}\big)$
  unless $\mathcal{P}$ fixes κ: $\kappa \leftarrow \max\{\kappa_{\min}, \kappa_{\text{run}}, \kappa_L\}$ ▷ (8); refuse if none exists
  $\kappa_{\text{run}} = \min\{k : (1 + e^{\varepsilon_{\text{run}}})\gamma(k) \le \zeta\delta_{\text{run}}\}$; $\kappa_L = \min\{k : \gamma(k) \le (\tilde{\rho}_{\text{run}}/\rho_L)\,\Gamma_L\}$
  $\gamma \leftarrow \gamma(\kappa)$; $\delta_{\text{DP}} \leftarrow \delta_{\text{run}} - (1 + e^{\varepsilon_{\text{run}}})\gamma$; **refuse if** $\delta_{\text{DP}} \le 0$
  $\rho_{\text{run}} \leftarrow \max\{\rho : \inf_{a>1} \frac{e^{(a-1)(a\rho - \varepsilon_{\text{run}})}}{a-1}\big(1 - \frac{1}{a}\big)^a \le \delta_{\text{DP}}\}$ ▷ tight zCDP conversion
  $\rho_1 \leftarrow f_1\rho_{\text{run}}$; $\rho_y \leftarrow f_y\rho_{\text{run}}$ (0 in joint mode); $\rho_2 \leftarrow \rho_{\text{run}} - \rho_1 - \rho_y$
  for Ω_f in increasing admissible scales **do** ▷ $\Omega_V = \Omega_f q_1 q_2/(\Omega_U \Omega_u)$, exactly representable
    for each release s: $\widehat{\Delta}_{\text{enc},s} \leftarrow \sqrt{2/N}\,\Omega_f\Delta_s + \Delta_{x,s} + \sqrt{N}$; $\sigma_{c,s}^2 \leftarrow \lceil \widehat{\Delta}_{\text{enc},s}^2/(2\rho_s) \rceil$
    if $\sigma_{c,s} \ge \max\{2^\kappa B_{\text{tot},s}(\Omega_f)/\sqrt{2\pi},\, 2^{20}\}$ for every s **then break** ▷ worst-case $B_{\text{tot},s}$
  end for ▷ refuse if no scale floods
  $\sigma_s \leftarrow \sigma_{c,s}\sqrt{N/2}/\Omega_f$; **refuse unless** $q_0 > 2\big(\Omega_f S_{\max} + 2^\kappa B_{\text{tot,max}} + t_\eta\sigma_{c,\max}\big)$
  with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$ ▷ no wraparound
  return κ, Ω_V, Ω_f, $\{\sigma_{c,s}^2, \sigma_s, \rho_s\}_s$ and $\gamma_s = \gamma/n_{\text{rel}}$ per release
end function

function SAMPLEDG(σ) exact discrete Gaussian on $\mathbb{Z}$ by rejection from a discrete Laplace, with big-integer
  arithmetic, a rational σ^2 and a CSPRNG; never by rounding a floating-point normal
end function

function SANITIZEBFV(ct) ▷ statistical flooding; BFV decryption stays exact
  $ct \leftarrow \mathsf{ModSwitch}_{q'}\big(ct \boxplus \mathsf{BFV.Enc}_{pk_B}(\mathbf{0})\big)$; $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}^{N_B}, \sigma_\psi}$ ▷ $\sigma_\psi \ge \max\{2^{\kappa_\psi}B_\psi/\sqrt{2\pi}, 2^{20}\}$
  return $ct \boxplus (\mathbf{e}, 0)$ ▷ requires $q'/(2t) > t_\eta\sigma_\psi + B_\psi$; B_ψ: Section 8
end function

function PREPROCESS(own columns, D_{ref}) ▷ each party separately; all constants public
  standardize each column with constants from D_{ref} or domain knowledge (O also standardizes y)
  optional tail transform per column: winsorize at $|z| \le c$, rank-Gaussian from the public quantiles, or log
  optional block normalization: scale the party's block of every row to unit norm (R without the intercept)
end function

function CHOOSEBOUNDS(D_{ref}) grid over quantile levels of $\|\dot{\mathbf{x}}\|$, $\|\mathbf{x}_O\|$ and $|y|$ in the preprocessed reference data; minimize
  the first-order coefficient MSE plus the squared clipping bias
end function

7 Output privacy

7.1 Sensitivity under replace-one adjacency

Stage 2 releases the stacked statistic

$$\mathbf{z}_2(D) = \big(\mathrm{triu}(\mathbf{B}),\ \mathbf{C},\ \mathbf{c}_R,\ \mathbf{c}_O,\ \mathbf{y}^\top\mathbf{y}\big), \tag{4}$$

without $\mathbf{y}^\top\mathbf{y}$ in separate mode, and stage 1 releases $\mathbf{z}_1(D) = (\mathrm{triu}\,\mathbf{B}, \mathbf{C})$. An unmatched row contributes nothing to these values. Replacing one matched row's $(\mathbf{x}, y)$ by $(\mathbf{x}', y')$, with R's row $\dot{\mathbf{x}}$ fixed, changes $\mathbf{z}_2$ by

$$\big(\mathrm{triu}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top),\ \dot{\mathbf{x}}(\mathbf{x}-\mathbf{x}')^\top,\ \dot{\mathbf{x}}(y-y'),\ \mathbf{x}y - \mathbf{x}'y',\ y^2 - y'^2\big).$$

Every entry of a release gets noise of the same s.d., so the relevant sensitivity is the ℓ_2 norm of this change, maximized over the bounds.

Proposition 7.1 (Replace-one sensitivity)**.** *With $r = B_R^2$, $o = B_O^2$, $w = B_y^2$ and $\Delta_{\mathrm{add}}^2 = (o+r)(o+w) + w^2$ the add/remove bound, $\Delta_{\mathrm{add}} \le \Delta \le 2\Delta_{\mathrm{add}}$, and*

$$\Delta^2 = \begin{cases} \tfrac{1}{2}(2o + r + w)^2 + 2rw & \text{if } |r - w| \le 2o, \\ 4r(o+w) & \text{if } r - w \ge 2o, \\ \max_{t\in[-B_y, B_y]} F\big(c^*(t), t\big) & \text{if } w - r > 2o, \end{cases} \tag{5}$$

where

$$F(c,t) = 2o^2(1-c^2) + 2ro(1-c) + r(B_y - t)^2 + o(w + t^2) - 2ocB_y t + (w - t^2)^2,$$
$$c^*(t) = \min\big\{1, \max\{-1, -(r + B_y t)/(2o)\}\big\}.$$

For stage 1, $\Delta_G = (2o + r)/\sqrt{2}$ if $r \le 2o$ and $2\sqrt{ro}$ otherwise. These are exact suprema for $d_O \ge 2$ and valid, conservative bounds for $d_O = 1$.

Proof sketch. The lower bound replaces $(\mathbf{x}, y)$ with $(\mathbf{0}, 0)$, the upper bound passes through that dataset with the triangle inequality. At the optimum $\|\dot{\mathbf{x}}\| = B_R$ and $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$; the remaining dependence is on $c = \cos\angle(\mathbf{x}, \mathbf{x}')$ and $t = y'$, and maximizing the concave quadratic in c gives $c^*(t)$. In regime 1 the worst case is $y' = -y$ with $c = -(r-w)/(2o)$ and a diagonal change in $\mathbf{B}$; in regime 2 it is $(\mathbf{x}, y) \mapsto (-\mathbf{x}, -y)$. Regime 3 reduces to the roots of three cubics. The full proof is in the companion proofs document (Part B, Theorem S1). □

[**Check 7:** DeltaRep in regime 3: the specification says one-dimensional maximization rounded up; the proofs give an exact enumeration over the roots of three cubics. Adopt the exact enumeration (keep rounding up as a safeguard)?]

Remark 7.1 (When $\mathbf{y}^\top\mathbf{y}$ is free)**.** Releasing $\mathbf{y}^\top\mathbf{y}$ costs nothing exactly when $\Delta = \Delta_{\neg yy}$. This holds whenever $w \le 2(r+o)$ and never when $w > 3(r+o)$; in between it is decided numerically. It is false in general: for $(B_R, B_O, B_y) = (1, 1, 5)$, $\Delta = 26.07$ with $\mathbf{y}^\top\mathbf{y}$ against 14.14 without. Algorithm 1 therefore releases $\mathbf{y}^\top\mathbf{y}$ in stage 2 only when it is free, and otherwise in its own release with sensitivity $\Delta_y = w$.

Remark 7.2 (Upper triangle or full matrix)**.** For $d_O \geq 2$ the supremum is the same whether the change in $\mathbf{B}$ is measured on its upper triangle or by the full Frobenius norm: the triu maximum is attained only when the change is diagonal. Releasing both triangles of $\mathbf{B}$ with independent noise would therefore cost the same worst-case ρ_2, and averaging the two copies would halve the variance of the off-diagonal entries.

[**Check 8:** Design choice raised by the remark above: release the full matrix B (both triangles, independent noise) instead of its upper triangle? Same worst-case privacy cost, lower variance off the diagonal; the bias formulas and the bootstrap would need the corresponding noise covariance.]

7.2 The release mechanism

For release s, let $\mathbf{z}_s(D) \in \mathbb{R}^{N_s}$ hold the released entries in their slots and zeros elsewhere.

Lemma 7.1 (Sensitivity of the encoded statistic)**.** $\|\tau_{\text{can}}^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\,\|\mathbf{z}\|_2$. *Hence* $\sup_{D\sim D'} \|\mathsf{Ecd}_\Omega(\mathbf{z}_s(D)) - \mathsf{Ecd}_\Omega(\mathbf{z}_s(D'))\|_2 \leq \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$, *where* $\sqrt{N}$ *bounds the difference of the two coordinate-wise roundings.*

The centre. The decrypted value differs from $\mathsf{Ecd}_{\Omega_f}(\mathbf{z}_s(D))$ also by deterministic, data-dependent terms, chiefly the spill of the one-hot masks. These terms are part of the query, not hints about secrets, so the mechanism is centred at the noiseless output of the circuit,

$$\hat{\mathbf{z}}_s(D) = \sum_{e\in\mathcal{E}_s} \frac{1}{q_1 q_2}\,\mathrm{Tr}\Big(\sum_{h\leq m} m_{U_e,h}\, m_{V_e,h}\Big) m_{u_e}, \qquad f_s(D) = \lfloor \hat{\mathbf{z}}_s(D) \rceil \in \mathbb{Z}^N,$$

where $m_{U_e,h}$, $m_{V_e,h}$ and m_{u_e} are the encodings of R's vectors at Ω_U, of O's vectors at Ω_V and of the mask at Ω_u, and Tr is the trace that EvalSum computes. $f_s(D)$ depends on the data and the public parameters only, not on any key, error or coin. Its sensitivity is (proofs, Part A, corollary on the sensitivity of the centre)

$$\sup_{D\sim D'} \|f_s(D) - f_s(D')\|_2 \leq \widehat{\Delta}_{\text{enc},s} := \sqrt{2/N}\,\Omega_f \Delta_s + \Delta_{x,s} + \sqrt{N}, \tag{6}$$

where $\Delta_{x,s}$, an explicit function of N, the encoding scales, q_1, q_2, $|\mathcal{E}_s|$, the bounds and Δ_s, bounds the change of the rounding and spill terms. Ω_f is the exact final scale, tracked from the primes and the encoding scales and never a nominal power of two, and R decodes with it. At the evaluation settings $\Delta_{x,2} \approx 2^{27.6}$ against $\sqrt{2/N}\,\Omega_f \Delta_2 \approx 2^{62.7}$, so centring changes ρ_s by a relative 2^{-34}. Flooding the mask spill instead would need $\log_2 \Omega_f$ of about 67–71 rather than 61.5 at the proofs' example ($\bar{n}_O = 10^6$, $\varepsilon = 1$).

Error decomposition. The value R reconstructs in a release is

$$v_s = h_O + c_1' s_R = c_0' + c_1' s + \eta_s = f_s(D) + e_{\text{tot},s} + \eta_s \pmod{q_0}.$$

$e_{\text{tot},s} \in \mathbb{Z}^N$ collects every term that depends on a key, an error or a coin: R's encryption errors multiplied by O's data, the rescaling and ModDown roundings, the key-switching error (which involves the errors of O's key shares), the mask spill of these errors, the rounding of the centre, and the fresh-encryption error $e\,u + e_0 + e_1 s$ of the re-randomization.

Lemma 7.2 (Smudging). *For $e \in \mathbb{Z}^N$ and $\eta \sim \mathcal{D}_{\mathbb{Z}^N,\sigma_c}$, $\mathrm{SD}(\eta + e, \eta) \le \|e\|_2/(2\sigma_c)$, and, for σ_c above the smoothing parameter, $\mathrm{SD}(\eta + e, \eta) \le \|e\|_2/(\sqrt{2\pi}\,\sigma_c) + \mathrm{negl}$. If $\sigma_c \ge \max\{2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi},\ 2^{20}\}$ and $\|e\|_2 \le B_{\mathrm{tot}}$, the distance is at most $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$, also given any side information independent of η.*

Theorem 7.1 (Worst-case error bound). *For the circuit of Algorithm 4 there is an explicit bound $B_{\mathrm{tot},s}$, a function of public parameters only (N, $\bar{n}_O$, d_R, d_O, B_R, B_O, B_y, the primes and the key-switching gadget, Ω_U, Ω_V, Ω_u, δ_B and the error distribution), such that $\Pr[\|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s}] \ge 1 - \delta_B$ for every dataset in the clipped domain, the probability being over keys and fresh randomness only (proofs, Part A, Theorem O1).*

At the proofs' example ($N = 2^{14}$, $\bar{n}_O = 10^6$, the bounds of the simulation population, all scales and both rescaling primes $2^{52.5}$, centred key switching, $\delta_B = 2^{-50}$) $B_{\mathrm{tot},2} \approx 2^{25.3}$, and the bound was checked against OpenFHE's noise estimates with a margin of 6–8 bits. With OpenFHE's key switching, whose fast base conversions are not centred, and the evaluation settings of Section 12, $B_{\mathrm{tot},2} \approx 2^{26.6}$.

CALIBRATE must use this worst-case, data-independent bound. A bound estimated from the actual ciphertexts would depend on the data, leak through $\sigma_{c,s}$, and not be a worst-case bound.

Theorem 7.2 (Output privacy against an honest-but-curious R). *Let the parameter vector satisfy the constraints of Section 12, and let* CALIBRATE*'s outputs, which depend on public values only, satisfy for every release s*

(i) $\sigma_{c,s} \ge \max\{2^\kappa B_{\mathrm{tot},s}/\sqrt{2\pi},\ 2^{20}\}$, *with $B_{\mathrm{tot},s}$ from Theorem 7.1;*

(ii) $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$ *and $\sum_s \rho_s \le \rho_{\mathrm{run}}$, where ρ_{run} is the largest ρ whose tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$, with*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big), \qquad \delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma > 0.$$

Assume RLWE for the re-randomizer (one sample per release, modulus q_0) and for O's key shares (multi-sample, with circular security for the rotation-key shares), at ring dimension N and modulus QP with $\log_2(QP) \le L_{\max}(N, \lambda_{\mathrm{sec}})$. Then R's view of one run is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP for $\sim$ with statistical slack γ, and therefore $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$-IND-CDP. The ideal mechanism $\mathcal{M}(D) = (f_s(D) + \eta_s)_s$ is a composition of lattice discrete Gaussian mechanisms and is ρ_{run}-zCDP. PSI security is not needed for this guarantee: the identifiers are fixed under $\sim$, so the PSI transcript does not depend on $(\mathbf{x}_O, y)$.

Proof sketch. A sequence of hybrids. H0 is the real view: setup messages, PSI messages, and $\{(c'_{1,s}, h_{O,s})\}_s$. H1 rewrites $h_{O,s} = v_s - c'_{1,s}s_R$, which is identical. H2 replaces v_s by $f_s(D) + \eta_s$; by Lemma 7.2 with side information and Theorem 7.1 this costs at most $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ per release, γ in total. After H2, $(u_s, e_{1,s})$ appear only through $c'_{1,s} = c_{1,s} + au_s + e_{1,s}$, because $c'_{0,s}$ is never sent. H3 replaces $c'_{1,s}$ by a uniform element (RLWE with secret u_s; fresh u_s per release). H4 simulates the setup messages from the RLWE assumption on O's key shares. The simulator, given $\mathcal{M}(D)$ and s_R, samples a uniform c^*_1 per release and outputs $h^* = \mathcal{M}(D)_s - c^*_1 s_R$. The multivariate discrete Gaussian with sensitivity $\widehat{\Delta}_{\mathrm{enc},s}$ is exactly ρ_s-zCDP [6]; costs add, and the tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP. Lemma 4.1 turns the slack into $\delta_{\mathrm{run}} = \delta_{\mathrm{DP}} + (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$. □

Remark 7.3 (Why the release format matters)**.** If O sent c'_0, or a share $c'_1 s_O$ without η, R would learn $c'_1 s_O$ exactly and could recover s_O, since c'_1 is invertible with overwhelming probability. Re-randomization matters too. The evaluated c_1 is computed from public values and O's plaintext columns, and R chose the seed of a, so without it c_1 could reveal O's data. In a real OpenFHE run of an earlier specification that added the noise as a plaintext scalar, without re-randomizing, R identified the true dataset in 20 of 20 trials, for a matched and for an unmatched record, where an $\varepsilon = 1$ guarantee caps the success rate at $e/(1+e) \approx 73\%$. Reusing η or u across releases would reveal $v_2 - v_1$ exactly, so both are fresh every time. Stage 1 uses its own packed ciphertext M_1, which depends only on triu $\mathbf{B}$ and $\mathbf{C}$, so its error cannot depend on entries released only in stage 2.

7.3 Calibration

CALIBRATE (Algorithm 7) derives every quantity the release needs from the parameter vector, using public values only, in five steps.

(1) *Mode and sensitivities.* Δ, $\Delta_{\neg yy}$, Δ_G and Δ_y from the bounds (Proposition 7.1). $\mathbf{y}^\top\mathbf{y}$ joins stage 2 iff it is free (Remark 7.1), so the number of releases is $n_{\mathrm{rel}} = 2$ (joint mode) or 3 (separate mode).

(2) *Flooding parameter and slack.* κ from (8) below, unless the parameters fix it; $\gamma = n_{\mathrm{rel}}(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97})$ and $\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$.

(3) *Budget.* ρ_{run}, the largest ρ whose tight conversion gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$, split as $\rho_1 = f_1\rho_{\mathrm{run}}$, $\rho_y = f_y\rho_{\mathrm{run}}$ in separate mode, and ρ_2 the rest.

(4) *Scale and noise.* The smallest admissible final scale Ω_f (that is, the smallest Ω_V, given Ω_U and Ω_u) at which $\sigma_{c,s} \geq \max\{2^\kappa B_{\mathrm{tot},s}(\Omega_f)/\sqrt{2\pi},\ 2^{20}\}$ for every release, with $\sigma^2_{c,s} = \lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s) \rceil$ an integer, as the exact sampler needs. Rounding up only lowers the cost below ρ_s.

(5) *Modulus and security.* $q_0 > 2(\Omega_f S_{\max} + 2^\kappa B_{\mathrm{tot,max}} + t_\eta\, \sigma_{c,\max})$ with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$, so that honest inputs never wrap around, and $\log_2(QP) \leq L_{\max}(N, \lambda_{\mathrm{sec}})$, the largest modulus at which RLWE with ring dimension N and ternary secrets has security λ_{sec} [1].

CALIBRATE refuses the run when a step has no solution: no admissible κ, $\delta_{\mathrm{DP}} \leq 0$, no scale that floods, q_0 too small, or a chain above the security bound.

The noise. The induced slot-domain noise is, up to a relative term $(\Delta_{x,s} + \sqrt{N})\sqrt{N/2}/(\Omega_f \Delta_s)$ that is below 10^{-10} at the evaluation settings,

$$\sigma_1 = \frac{\Delta_G}{\sqrt{2 f_1 \rho_{\mathrm{run}}}}, \qquad \sigma_2 = \frac{\Delta}{\sqrt{2(1-f_1)\rho_{\mathrm{run}}}}, \qquad \sqrt{\rho_{\mathrm{run}}} \gtrsim \sqrt{L + \varepsilon_{\mathrm{run}}} - \sqrt{L}, \quad L = \ln(1/\delta_{\mathrm{DP}}), \tag{7}$$

in joint mode ($\Delta_{\neg yy}$ and $1 - f_1 - f_y$ replace Δ and $1 - f_1$ in separate mode). The last inequality, from the simple conversion, gives a planning formula; over $\varepsilon \in [0.1, 8]$ and $\delta \in [10^{-9}, 10^{-6}]$ it is conservative by 6–33% in σ_2 (Section 13.2).

The flooding parameter. The same η is the DP noise and floods $e_{\mathrm{tot},s}$, and the flooding enters the guarantee only through the slack γ, which Lemma 4.1 converts with the factor

$1 + e^{\varepsilon}$. The flooding parameter is therefore a function of the budget, not a constant. Unless the parameters fix it, CALIBRATE sets

$$\kappa = \max\{\kappa_{\min},\ \kappa_{\text{run}},\ \kappa_L\}, \tag{8}$$

where κ_{run} is the smallest integer with $(1 + e^{\varepsilon_{\text{run}}})\gamma \le \zeta\,\delta_{\text{run}}$, and κ_L is the smallest integer with $\gamma \le (\tilde{\rho}_{\text{run}}/\rho_L)\,\Gamma_L$, this run's share of the ledger's slack capacity (Section 7.4); $\tilde{\rho}_{\text{run}}$ is the budget at $\delta_{\text{DP}} = (1-\zeta)\delta_{\text{run}}$, a lower bound on ρ_{run} once $\kappa \ge \kappa_{\text{run}}$. The three terms have separate jobs.

- κ_{run} makes the per-run statement hold with the slack taking at most a share ζ of δ_{run}. γ falls with κ to the floor $n_{\text{rel}}(\delta_B + 2^{-97})$, so κ_{run} exists iff $n_{\text{rel}}(1 + e^{\varepsilon_{\text{run}}})(\delta_B + 2^{-97}) < \zeta\delta_{\text{run}}$; otherwise δ_B must be lowered.
- κ_L makes the ledger run out of zCDP budget before it runs out of slack, whatever the order and the sizes of the runs.
- $\kappa_{\min}$ is a statistical security floor. γ also bounds what R can learn beyond $\mathcal{M}(D)$, including about O's key share, so it should not grow with a generous δ_{run}.

ζ trades budget for bits. Shrinking δ_{DP} from δ_{run} to $(1-\zeta)\delta_{\text{run}}$ lowers ρ_{run} by a relative $\zeta/\sqrt{L(L+\varepsilon_{\text{run}})}$ to first order (simple conversion, $L = \ln(1/\delta_{\text{run}})$), so σ_2 rises by half that, while each unit of κ adds one bit to Ω_f and to q_0. At $\delta_{\text{run}} = 10^{-9}$ and $\varepsilon_{\text{run}} \in \{1, 4, 8\}$ the tight conversion loses 0.46–0.59% of ρ_{run} at $\zeta = 0.1$ and 0.04–0.06% at $\zeta = 0.01$, and $\zeta = 0.01$ costs 3–5 more bits of κ. Table 4 gives κ and the resulting scale at the evaluation settings.

Table 4: Example at the evaluation settings: the flooding parameter from (8) and the final scale CALIBRATE chooses, simulation population ($p = 11$, joint mode, $n_{\text{rel}} = 2$), $\delta_{\text{run}} = 10^{-9}$, $\zeta = 0.1$, $\kappa_{\min} = 40$, $\Omega_U = 2^{68}$, $N = 2^{14}$, $q_0 \approx 2^{119}$, $\log_2(QP) = 357 \le 438$. Left: a single-run ledger. Right: ledger cap $(\varepsilon_L, \delta_L) = (10, 10^{-6})$ with $\zeta_L = 0.1$. Every row is feasible. At a fixed $\kappa = 40$, $\varepsilon_{\text{run}} = 8$ is not: the slack exceeds δ_{run}.

	single-run ledger						cap $(10, 10^{-6})$		
ε_{run}	κ_{run}	κ	slack (% of δ)	$\log_2 \Omega_f$ $\bar{n}_O = 4{,}096$	10^6	$\log_2 q_0$ needed, 10^6	κ_L	κ	$\log_2 \Omega_f$ 10^6
0.5	36	40	0.48	61.78	61.78	88.0	48	48	69.88
1	37	40	0.68	62.75	62.75	89.0	46	46	68.78
2	38	40	1.53	63.69	63.69	89.9	44	44	67.72
4	41	41	5.07	65.62	65.66	91.9	42	42	66.66
8	46	46	9.00	71.56	71.94	98.2	40	46	71.94

The scale. Since $\sigma_{c,s} = \sigma_s \Omega_f \sqrt{2/N}$ and σ_s is fixed by the budget, O meets the flooding condition by raising Ω_f; stage 2 binds:

$$\log_2 \Omega_f \approx \kappa + \log_2 B_{\text{tot},2} + \log_2 \sqrt{N/2} - \log_2 \sqrt{2\pi} - \log_2 \sigma_2.$$

Example: for the simulation population at $\varepsilon = 1$, $\delta = 10^{-9}$, $N = 2^{14}$ and $\kappa = 40$, the proofs' bound gives about $40 + 25.3 + 6.5 - 1.3 - 9.0 \approx 61.5$, and the bound for OpenFHE's key switching 62.75 (Table 4).

Encoding scales. We use the circuit as written (Algorithm 4) with separate encoding scales: Ω_U and Ω_u are parameters, CALIBRATE chooses Ω_V, and $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1q_2)$ follows and is tracked exactly. The final scale is therefore not tied to the size of a rescaling prime, which a 64-bit implementation keeps below 2^{60}. The constraint on Ω_U comes from B_{tot}: the term from R's encryption errors grows like $(\Omega_f/\Omega_U)\sqrt{\bar{n}_O}$. With $\Omega_U \geq \Omega_f$ it stays below the other terms at every register size; with $\Omega_U < \Omega_f$ the required scale grows with $\bar{n}_O$, and for large enough κ no scale floods, because B_{tot} then grows as fast as Ω_f, and CALIBRATE refuses. In the evaluation $\Omega_U = 2^{68}$ and $\Omega_u = q_1$. The required scale is then independent of $\bar{n}_O \in [4096, 10^6]$ to within 0.05 bits for $\varepsilon \leq 4$, and grows by 0.4 bits at $\varepsilon = 8$, where $\Omega_f > \Omega_U$ (Table 4). With $\Omega_U = q_2 \approx 2^{59}$ instead, the required scale reaches $2^{70.3}$ at $\bar{n}_O = 10^6$, $\varepsilon = 4$ and $\kappa = 40$. The proof review's restructured circuit, which needs about $2^{53.7}$, is not used: its proof is incomplete, and with decoupled scales the larger scale costs nothing.

7.4 The ledger and composition across runs

O keeps one ledger per base register, shared across all researchers and scopes, and refuses to produce a decryption share once the cap would be exceeded. Because no decryption is possible without O's share, the cap is enforced technically, not by policy. A refused stage-2 gate (Section 9.3) still spends the budget. The OPRF quota limits membership look-ups across sessions as a matter of policy.

Parameters and state. The ledger's parameters are a cap $(\varepsilon_L, \delta_L)$ and the share $\zeta_L \in (0,1)$ of δ_L reserved for statistical slack. From them, ρ_L is the largest ρ whose tight conversion gives $(\varepsilon_L, (1-\zeta_L)\delta_L)$, and $\Gamma_L = \zeta_L\delta_L/(1+e^{\varepsilon_L})$. The state is the sum ρ_Σ of the zCDP costs ρ_s and the sum Γ_Σ of the raw slacks $\gamma_s = \delta_B + 2^{-\kappa}(1+2^{-80}) + 2^{-97}$ of every release made on the register. The raw slacks add, not the converted ones $(1+e^{\varepsilon_{\text{run}}})\gamma$: they are converted once, at the cap's ε_L.

Admission. Write $\delta(\rho, \varepsilon)$ for the tight conversion.

- *Runs fixed in advance* (non-adaptive). Admit a run iff $\delta(\rho_\Sigma + \rho_{\text{run}}, \varepsilon_L) + (1+e^{\varepsilon_L})(\Gamma_\Sigma + \gamma) \leq \delta_L$.
- *Runs chosen after seeing earlier results* (adaptive). Admit a run iff $\rho_\Sigma + \rho_{\text{run}} \leq \rho_L$ and $\Gamma_\Sigma + \gamma \leq \Gamma_L$. This is a privacy filter in the sense of Rogers et al. [28], and it is the rule for a shared register, where later runs are rarely fixed in advance. With κ_L in (8) the slack condition holds whenever the zCDP condition does.

A register on which a single run is allowed is the special case $(\varepsilon_L, \delta_L) = (\varepsilon_{\text{run}}, \delta_{\text{run}})$ under the first rule, which that run fills exactly; then $\kappa_L = \kappa_{\text{run}}$.

Proposition 7.2 (Composition across runs)**.** *Let runs $k = 1, 2, \ldots$ on one base register use independent keys, shuffles and randomness, let each satisfy the hypotheses of Theorem 7.2 with zCDP cost ρ_k and slack γ_k, and let the number of runs be polynomial in λ_{sec}. If the runs are fixed in advance and admitted by the first rule, or chosen adaptively and admitted by the second, then the joint view of all researchers is $(\varepsilon_L, \delta_L)$-IND-CDP for $\sim$.*

Proof sketch. Simulation, run by run. Replace the runs one at a time by their simulations $\mathcal{S}_k(\mathcal{M}_k(D))$. Run k uses keys and randomness independent of the other runs, so the reduction

for each hybrid step can generate every other run, real or simulated, from D. This holds also when the parameters of run k are a function of the researchers' views of earlier runs, since the reduction can compute that function. Step k costs the expectation of γ_k plus a negligible term, so the joint view is simulated from $(\mathcal{M}_k(D))_k$ with slack $\Gamma = \mathbb{E}\sum_k \gamma_k$, which is at most Γ_Σ for fixed runs and at most Γ_L under the filter. For fixed runs this is the composition lemma of the proofs (Part A); the adaptive case is the same hybrid. *The ideal mechanisms.* For fixed runs, zCDP costs add [5]. For adaptive runs, $\mathcal{M}_k$ is ρ_k-zCDP, that is, $(a, a\rho_k)$-Rényi DP for every order $a > 1$, with ρ_k a function of earlier outputs and $\sum_k \rho_k \le \rho_L$ always. The Rényi filter of Feldman and Zrnic [13] (Theorem 1.1), applied at each order a, makes the composition $(a, a\rho_L)$-Rényi DP for every a, that is, ρ_L-zCDP. *Conversion.* The tight conversion gives $(\varepsilon_L, \delta(\rho, \varepsilon_L))$-DP for the composed mechanism, and Lemma 4.1 adds $(1 + e^{\varepsilon_L})\Gamma$; each admission rule keeps the sum within δ_L. □

[**Check 9:** Composition: the adaptive hybrid of Proposition 7.2 is argued here but not yet written out in the proofs, which prove the non-adaptive case. The literature was checked only through search-engine extracts, not the full texts: Feldman and Zrnic's Theorem 1.1 as quoted from the arXiv version (check its number in the NeurIPS version, and the page numbers); the pages of Rogers et al. (1921–1929); the volume and pages of Whitehouse et al. (PMLR 202, 36990–37007).]

Two consequences. First, a statement over many runs converts the slack with $1 + e^{\varepsilon_L}$, not $1 + e^{\varepsilon_{\text{run}}}$, so a ledger with a large ε_L needs a larger κ per run. At $(\varepsilon_L, \delta_L, \zeta_L) = (10, 10^{-6}, 0.1)$, for example, κ_L is 48, 46, 44, 42 and 40 for $\varepsilon_{\text{run}} = 0.5, 1, 2, 4, 8$ (Table 4); at $\varepsilon_L = 20$ the floor $n_{\text{rel}}\delta_B$ alone exceeds every run's share unless δ_B is lowered well below 2^{-50}. Second, the filter loses nothing against non-adaptive composition, because zCDP composes by summing a single parameter. For (ε, δ)-DP, filters that match advanced composition, and a filter for approximate zCDP, are given by Whitehouse et al. [31].

Remark 7.4 (A sharper slack)**.** The factor $1 + e^{\varepsilon_L}$ comes from measuring the smudging step of Theorem 7.2 in statistical distance. A shifted discrete Gaussian $\eta + e$ with $\|e\|_2 \le B_{\text{tot}}$ is also (ε', δ')-indistinguishable from η with ε' of order $2^{-\kappa}\sqrt{\ln(1/\delta')}$. Measuring that step in hockey-stick divergence would move most of the slack into a negligible ε' and leave only δ_B and a small δ' to be multiplied by e^{ε_L}, which would lower κ_L substantially. It needs a new proof of the hybrids and is not used here.

8 Input privacy, cohort privacy and correctness

G1. O sees R's seeded uploads $c_0 = -as_R + e + \mathsf{Ecd}(\mathbf{v})$, which are pseudorandom under RLWE in s_R (with circular security covering R's rotation-key shares, and the PRG modelled as a random oracle); the PSI messages, which are BFV ciphertexts under R's own key and blinded OPRF elements; and R's key-generation shares, which are RLWE samples. O never receives a decryption share from R, so its view can be simulated from $(n_R^{\max}, \bar{n}_O)$, up to the failure probability of cuckoo hashing.

G2. O evaluates exactly $n_R^{\max}$ OPRF calls on blinded inputs and answers padded PSI queries, so it learns nothing about the cohort or n_R.

G4. R learns I and $\pi|_I$ from linkage. That it learns nothing else about O's register needs the labeled PSI to be secure against a semi-honest receiver. Chen, Laine and Rindal prove their PSI from homomorphic encryption secure in the semi-honest model assuming the scheme is IND-CPA secure and circuit private, and note that statistical function privacy, obtained in the leveled setting by re-randomization and noise flooding, suffices [7]. Chen, Huang, Laine and Rindal add OPRF preprocessing, which strengthens the security model [8]. Two layers of the protocol address what the noise of a reply could reveal.

(a) *OPRF-keyed items and labels.* Every item O inserts is $F_k(\mathrm{id}_j)$ under a fresh key, and every label is $\tau(j)$ masked by a key derived from $F'_k(\mathrm{id}_j)$. R learns F_k only at its $n_R^{\max}$ queries, so the item and the label of every row outside I are pseudorandom to it; τ is a uniform permutation, and the masks $\mathbf{r}_1$, $\mathbf{r}_2$ are fresh. A reply, noise included, is a deterministic function of R's ciphertexts and of these values. A simulator that knows $(I, \pi|_I)$, the public sizes and R's view of the OPRF can therefore run O's evaluation on fresh random items and labels outside I and obtain replies indistinguishable from the real ones, under the security of the OPRF. Even without circuit privacy, then, what the noise reveals about O's register beyond $(I, \pi|_I)$ is pseudorandom.

(b) SanitizeBFV (Algorithm 7) is part of the protocol nevertheless, so that G4 does not rest on the pseudorandomness of the evaluation's inputs alone. It switches each reply to the reply modulus q', adds a fresh encryption of zero under R's BFV key, and floods the noise with an exact discrete Gaussian $\mathcal{D}_{\mathbb{Z}^{N_B},\sigma_\psi}$. Let B_ψ bound, with probability at least $1-\delta_{B,\psi}$ and for every input of O, the ℓ_2 norm of the switched reply's error other than the flooding noise. The requirements are

$$\sigma_\psi \geq \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi},\ 2^{20}\}\ \text{(flooding)}, \qquad q'/(2t) > t_\eta\, \sigma_\psi + B_\psi\ \text{(exact decryption)}.$$

Then, as in hybrids H2 and H3 of Theorem 7.2, each reply ciphertext is within statistical distance $\delta_{B,\psi}+2^{-\kappa_\psi}(1+2^{-80})+2^{-97}$, plus a computational term (RLWE in the randomness of the fresh encryption), of a ciphertext computed from its plaintext and fresh randomness alone. Over the n_ψ reply ciphertexts of a run the distances add. The plaintexts themselves are 0 or uniform, and a label or uniform, because of the masks.

Neither layer is charged to δ: the PSI transcript does not depend on $(\mathbf{x}_O, y)$, so it lies outside the DP guarantee, and κ_ψ is a security parameter of G4 alone. A modulus switch alone, as in the specification, is heuristic. Flooding protects against a semi-honest receiver, whose ciphertexts are fresh; a malicious receiver could send noisier ones, which is outside our model.

At the evaluation settings the implementation uses $\kappa_\psi = 40$, $\sigma_\psi = 2^{\kappa_\psi} B_\psi \approx 2^{61}$ with $B_\psi = 2(\sqrt{N_B}/2 + N_B^{3/2}/2) + 1 \approx 2^{21}$, and q' of two 60-bit limbs, so $q'/(2t) \approx 2^{87}$ against $10\,\sigma_\psi \approx 2^{64.3}$. Both requirements hold for the rounding terms of the switch, which this B_ψ bounds in the worst case for a ternary key. The switched evaluation noise is covered by the $+1$ only by measurement (2^{-72} after the switch), not by a worst-case bound.

[**Check 10:** G4: derive a worst-case, data-independent bound on the BFV evaluation noise of the PSI replies after the switch to q', so that B-psi is proven (the implementation's value is measured). The circuit-privacy statements of CLR17 and the role of OPRF preprocessing in CHLR18 were checked only through search-engine extracts; confirm against the full texts, including whether CHLR18 state that OPRF preprocessing removes the need for circuit privacy.]

Correctness. With q_0 from CALIBRATE, honest inputs never wrap around; the failure probability is at most $\delta_B + 2Ne^{-t_\eta^2/2}$ per release, a correctness event that is not charged to δ. With the noise switched off, decryption reproduces the plaintext statistics up to the CKKS approximation error (Section 13.1).

9 The statistical procedure

9.1 Estimands

The protocol targets two quantities.

- The **ridge target** $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$, a fixed function of the matched data once (λ, Ψ) is chosen. Only the mechanism noise enters its uncertainty.
- The **regression coefficient** $\boldsymbol{\beta}$ of the working linear model for the clipped data, whose sample analogue is $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$. Its uncertainty includes sampling variance, and $\boldsymbol{\theta}_\lambda$ is biased for it when $\lambda > 0$.

Either the working model (conditionally independent, homoskedastic errors) is assumed for the clipped data—it holds if the bounds are true bounds, so that clipping never binds—or $\boldsymbol{\beta}$ is the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top \mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$ of the clipped data. In the second case the bias bound below remains valid but the homoskedastic variance does not, because clipping induces heteroskedasticity.

[**Check 11:** Choose the definition of beta (working model on clipped data, or projection coefficient) and say what the bias-aware interval guarantees under heteroskedasticity, which is still open.]

9.2 Stage 1: choosing the ridge

Stage 1 releases $\hat{\mathbf{B}}_1$ and $\hat{\mathbf{C}}_1$ with s.d. σ_1, and R forms $\hat{\mathbf{G}}_1$ with its exact $\mathbf{A}$. With $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)})$,

$$\Pr\left[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1\right] \le \alpha_L,$$

so $\hat{L} = \max\{0, \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1\} \le \lambda_{\min}(\mathbf{G})$ with probability at least $1 - \alpha_L$. This one-sided event is the only one the procedure uses; a two-sided statement needs $\ln(2/\alpha_L)$. At $p = 11$ and $\alpha_L = 0.0025$ (the evaluation setting) the radius is about 1.55 times conservative.

[**Check 12:** r-1: optionally set it to the exact (1 - alpha-L) quantile of lambda-max of the noise, computed offline by Monte Carlo (7.43 sigma-1 instead of 11.53 sigma-1 for p = 11), at no privacy cost.]

With the target $g^\star = 2\vartheta^\star\sigma_2\sqrt{p}$, the full ridge

$$\lambda_I = \max\{0,\ g^\star + r_1 - \lambda_{\min}(\hat{\mathbf{G}}_1)\}$$

guarantees $\lambda_{\min}(\mathbf{G} + \lambda_I\mathbf{I}) \ge g^\star$ on that event. The O-block ridge λ_O is the smallest λ with $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \ge g^\star + r_1$. It is finite iff $\lambda_{\min}(\mathbf{A}) > g^\star + r_1$, and the Π_O ridge can never certify a validity ratio above $\lambda_{\min}(\mathbf{A})/(2\sigma_2\sqrt{p})$. The rule takes $(\lambda, \Psi) = (\lambda_O, \Pi_O)$ if $\lambda_O \le \kappa_\Psi\lambda_I$, for a parameter $\kappa_\Psi \ge 0$, and $(\lambda_I, \mathbf{I})$ otherwise. Shrinking only O's block leaves R's coefficients unshrunk when R's own block is well conditioned.

Proposition 9.1 (Independence)**.** (λ, Ψ) *is a function of the stage-1 release and of* $\mathbf{A}$ *only, and is therefore independent of the stage-2 noise.*

This is what makes the bias theory of Section 10 and the intervals below valid conditionally on (λ, Ψ). Choosing λ from the same release that is used for estimation would make λ a function of the noise; Section 10.5 shows what goes wrong.

[**Check 13:** The evaluation settings f-1 = 0.1, theta-star = 2.5 and kappa-Psi = 2 were chosen by simulation of earlier variants; confirm them under the two-stage rule.]

9.3 Stage 2: estimate and gate

Stage 2 releases every entry with s.d. σ_2. R assembles $\hat{\mathbf{G}}_2$ and $\hat{\mathbf{c}}_2$, computes the validity ratio $\hat{\vartheta} = \lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)/(2\sigma_2\sqrt{p})$, and releases nothing if $\hat{\vartheta} < 1$. Otherwise

$$\tilde{\boldsymbol{\theta}} = \tilde{\mathbf{P}}\hat{\mathbf{c}}_2, \qquad \tilde{\mathbf{P}} = (\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1}.$$

The gate is post-processing; a refused release still spends the budget. For $\vartheta^\star > 2$ the probability of refusal is at most $\alpha_L + \exp(-p(\vartheta^\star - 2)^2)$ (proofs, Part B, Lemma S3).

[**Check 14:** Say whether the coverage statements below are conditional on passing the gate.]

9.4 Intervals

Mechanism variance. R knows the noise distribution, so it estimates the covariance of $\tilde{\boldsymbol{\theta}}$ due to the mechanism by a parametric bootstrap around its own release: symmetric noise $\mathbf{E}^*$ with s.d. σ_2 on the O-dependent entries of $\hat{\mathbf{G}}_2$ and zero on $\mathbf{A}$, noise $\mathbf{f}^* \sim \mathcal{N}(0, \sigma_2^2\mathbf{I})$ on $\hat{\mathbf{c}}_2$, and $\boldsymbol{\theta}_b^* = (\hat{\mathbf{G}}_2 + \mathbf{E}_b^* + \lambda\Psi)^{-1}(\hat{\mathbf{c}}_2 + \mathbf{f}_b^*)$. A draw is kept only if $\|\tilde{\mathbf{P}}^{1/2}\mathbf{E}^*\tilde{\mathbf{P}}^{1/2}\|_{\mathrm{op}} \le \eta_B$, for a truncation parameter $\eta_B \in (0,1)$; without truncation the bootstrap moments do not exist. With B draws the Monte Carlo error of $\mathbf{V}_{\mathrm{mech}}$ is of relative order $B^{-1/2}$. Antithetic pairs $(\pm\mathbf{E}^*, \pm\mathbf{f}^*)$ reduce Monte Carlo error. $\mathbf{V}_{\mathrm{mech}}$ is the sample covariance of the kept draws. To first order, $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda \approx \tilde{\mathbf{P}}(\mathbf{f} - \mathbf{E}\boldsymbol{\theta}_\lambda)$, so

$$\mathbf{V}_{\mathrm{mech}} \approx \sigma_2^2\,\tilde{\mathbf{P}}\big(\mathbf{I} + K(\tilde{\boldsymbol{\theta}})\big)\tilde{\mathbf{P}}, \qquad K(\boldsymbol{\theta})_{ii} = \textstyle\sum_j \chi_{ij}\theta_j^2, \quad K(\boldsymbol{\theta})_{ik} = \chi_{ik}\theta_i\theta_k \ (i \neq k),$$

with $\chi_{ij} = 0$ iff i and j both lie in R's block. R may use this closed form instead of the bootstrap.

Wald interval for $\boldsymbol{\theta}_\lambda$. The error level is split as $\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$, a parameter (the evaluation setting is $\alpha = 0.05$ with $\alpha_L = \alpha_U = \alpha/20$, so $\alpha_{\mathrm{CI}} = 0.045$ and $z_{1-\alpha_{\mathrm{CI}}/2} = 2.005$). The interval is

$$\tilde{\theta}_j \pm z_{1-\alpha_{\mathrm{CI}}/2}\sqrt{(\mathbf{V}_{\mathrm{mech}})_{jj}}.$$

Residual scale. $\widetilde{\mathrm{RSS}} = \widehat{yy} - \hat{\mathbf{c}}_2^\top\tilde{\boldsymbol{\theta}} - \lambda\tilde{\boldsymbol{\theta}}^\top\Psi\tilde{\boldsymbol{\theta}}$ is biased by the noise at order σ_2^2. The same bootstrap gives $\mathrm{Corr} = \overline{\mathrm{RSS}^*} - \widetilde{\mathrm{RSS}}$ and $\tilde{s}^2 = \max\{\widetilde{\mathrm{RSS}} - \mathrm{Corr}, 0\}/(n-p)$, which removes the $O(\sigma_2^2)$ bias up to $O(\sigma_2^4)$.

Bias-aware interval for $\boldsymbol{\beta}$. The total variance adds the sampling term, $\mathbf{V} = \mathbf{V}_{\mathrm{mech}} + \tilde{s}^2\,\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$. The ridge bias of coordinate j is at most $\bar{b}_j = \lambda\|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|_2\,\hat{B}_\beta$, where $\hat{B}_\beta = \min\{B_\beta^{\mathrm{pub}}, \sqrt{U_{yy}/\hat{L}}\}$ bounds $\|\boldsymbol{\beta}\|$ from a public a-priori bound and a private one,

$$U_{yy} = \min\Big\{\widehat{yy} + z_{1-\alpha_U/2}\,\sigma_{yy} + B_y^2\sqrt{n\ln(2/\alpha_U)/2},\ nB_y^2\Big\},$$

with σ_{yy} the s.d. of the release that carries $\mathbf{y}^\top\mathbf{y}$. The interval, following Armstrong, Kolesár and Kwon [2], is

$$\tilde{\theta}_j \pm \mathrm{cv}_{\alpha_{\mathrm{CI}}}\big(\bar{b}_j/\sqrt{V_{jj}}\big)\sqrt{V_{jj}},$$

where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$ quantile of $|Z+t|$ for $Z\sim\mathcal{N}(0,1)$. The private bound is usually infinite while the ridge is on ($\hat{L}=0$), so without a public B_β^{pub} the interval for $\boldsymbol{\beta}$ is unbounded exactly when the ridge is needed. This is the honest answer: the data cannot rule out large coefficients in the directions the ridge suppresses.

Remark 9.1 (How accurate is first order?)**.** The validity ratio controls convergence of the Neumann series, but the relative size of the neglected mechanism bias in the Wald interval is of order $\sigma_2\,\mathrm{tr}(\mathbf{P}_\lambda) \le \sqrt{p}/(2\vartheta_0)$, which at $\vartheta^\star = 2.5$ and $p = 11$ (the evaluation setting) can be 0.66. Coverage claims are therefore asymptotic as $\vartheta_0 \to \infty$ at fixed p. Algorithm 6 reports $\sigma_2\,\mathrm{tr}(\tilde{\mathbf{P}})$ as a diagnostic.

[**Check 15:** Consider centring the intervals at the bootstrap bias-corrected estimate 2 theta-tilde minus the bootstrap mean, as the proof review suggests.]

9.5 When can the ridge be dropped?

The stage-1 rule is explicit, so the intersection size from which it stops adding a ridge can be read off the protocol parameters and one property of the data.

The threshold. Let $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ be the per-record second-moment matrix of the preprocessed, clipped design, intercept included, and $\ell = \lambda_{\min}(\Sigma)$ its weakest direction. With n matched records $\mathbf{G} \approx n\Sigma$ and $\lambda_{\min}(\mathbf{G}) \approx n\ell$, up to a relative fluctuation of order $n^{-1/2}$. Ignoring the stage-1 noise in $\lambda_{\min}(\hat{\mathbf{G}}_1)$, the full ridge is $\lambda_I(n) \approx \max\{0,\ g^\star + r_1 - n\ell\}$, which falls linearly in n and vanishes at

$$n_{\mathrm{off}} = \frac{g^\star + r_1}{\ell} = \frac{\sqrt{p}}{\ell\sqrt{2\rho_{\mathrm{run}}}}\left[\frac{2\vartheta^\star\Delta}{\sqrt{1-f_1}} + \Big(2 + 2\sqrt{\ln(1/\alpha_L)/p}\Big)\frac{\Delta_G}{\sqrt{f_1}}\right], \tag{9}$$

by (7). At the evaluation settings ($\vartheta^\star = 2.5$, $f_1 = 0.1$, $\alpha_L = 0.0025$) and $p = 11$ this is

$$n_{\mathrm{off}} \approx \frac{17.5\,\Delta + 36.5\,\Delta_G}{\ell\sqrt{2\rho_{\mathrm{run}}}},$$

with $\sqrt{2\rho_{\mathrm{run}}} = 0.089$, 0.173, 0.335, 0.642 and 1.21 for $\varepsilon = 0.5$, 1, 2, 4 and 8 at $\delta = 10^{-9}$. Since $\sqrt{\rho_{\mathrm{run}}}$ grows almost linearly in ε at fixed δ, n_{off} scales like the sensitivity divided by $\ell\varepsilon$. The two terms have different origins. $g^\star/\ell$ is the threshold if ℓ were known in advance; r_1/ℓ is the price of learning it from stage 1. With $\Delta_G \approx 0.9\Delta$, as for the simulation population, $r_1 \approx 1.9g^\star$, so learning ℓ multiplies the threshold by about 2.9. The private bound $\hat{L}$ on $\lambda_{\min}(\mathbf{G})$, which makes the interval for $\boldsymbol{\beta}$ finite without a public bound on $\|\boldsymbol{\beta}\|$, becomes positive earlier, from $n \approx r_1/\ell$ (about $0.65\,n_{\mathrm{off}}$ there).

Below the threshold. With $\Psi = \mathbf{I}$, the component of $\boldsymbol{\theta}_\lambda$ along an eigenvector of $\mathbf{G}$ with eigenvalue $n\ell_k$ keeps the fraction $n\ell_k/(n\ell_k + \lambda)$ of the least-squares coefficient. In the weakest direction $n\ell + \lambda \approx g^\star + r_1 = n_{\mathrm{off}}\ell$, so the fraction kept is n/n_{off}: the shrinkage there falls linearly in n and is gone at n_{off}. Directions with $\ell_k \gg \ell$ are barely shrunk. A weak direction carries little of the variation in $\mathbf{x}$, so shrinking it costs little in prediction error but a lot in the coefficients that load on it; this is why prediction recovers before the coefficients do (Section 13.4).

The transition. The noise lowers $\lambda_{\min}(\hat{\mathbf{G}}_1)$ on average, the more so when several eigenvalues of Σ are close to ℓ, so the rule chooses a slightly larger λ than the formula and switches off slightly after n_{off}. Table 5 simulates the stage-1 rule on the simulation population of Section 13.5 ($\ell = 0.598$, $\Delta = 86.2$, $\Delta_G = 77.2$, $p = 11$): the ridge is never off at $0.8\,n_{\text{off}}$, off in 1–3% of releases at n_{off} and in 96–98% at $1.2\,n_{\text{off}}$. We take $1.2\,n_{\text{off}}$ as the size from which the ridge is off with high probability. Table 13 uses the committed bound $\ell = 0.494$ instead of the population value, so its thresholds are 21% higher. Under the earlier mechanism, which used that committed bound instead of a stage-1 release, the ridge switched off at exactly $g^\star/\ell$ for that mechanism's σ and $\vartheta^\star = 2$ (Section 13.5).

Table 5: Ridge threshold for the simulation population, $\delta = 10^{-9}$, evaluation settings. n_{off} from (9); $g^\star/\ell$ is the threshold with ℓ known in advance. Last three columns: fraction of 400 simulated stage-1 releases with $\lambda = 0$ at $n = 0.8$, 1.0 and 1.2 times n_{off}.

ε	σ_1	σ_2	$g^\star$	r_1	n_{off}	$g^\star/\ell$	0.8	1.0	1.2
0.5	2746	1022	16942	31654	81,280	28,337	0.00	0.03	0.97
1	1411	525	8705	16265	41,764	14,560	0.00	0.03	0.96
2	729	271	4496	8400	21,570	7,520	0.00	0.01	0.98
4	380	141	2345	4382	11,251	3,922	0.00	0.03	0.97
8	202	75	1244	2325	5,970	2,081	0.00	0.03	0.98

Check on real data. On protein (Section 13.4, 19,206 matched records, $p = 10$), ℓ computed from the matched design predicts the ridge the encrypted runs chose (Table 6). The runs chose 1–19% more than predicted, always more, as the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ implies; the gap is largest where σ_1 is largest.

Table 6: Protein, 19,206 matched records, $\delta = 10^{-9}$: ℓ of the matched design, n_{off} from (9), the ridge $g^\star + r_1 - n\ell$ it predicts, and the median ridge of the five encrypted runs of Table 15.

Preprocessing	Δ	Δ_G	ℓ	ε	n_{off}	predicted λ	encrypted λ
standardize	87.3	80.9	0.00191	1	13.1 M	24,953	29,678
standardize	87.3	80.9	0.00191	4	3.53 M	6,695	7,321
winsorize $\lvert z\rvert \le 2$	40.5	34.2	0.00125	1	8.74 M	10,902	12,552
winsorize $\lvert z\rvert \le 2$	40.5	34.2	0.00125	4	2.35 M	2,919	3,088
block unit norm	8.4	2.83	0.00201	1	691 k	1,353	1,385
block unit norm	8.4	2.83	0.00201	4	186 k	336	338

What sets ℓ. ℓ is a property of the joint design after preprocessing, so near-collinearity decides n_{off}, and it may lie within either party's block or across them. On protein, ℓ is about 0.002 under every preprocessing, which keeps the ridge on to between 186k and 13M records. Its source differs.

- Standardized: the weakest direction is $0.77\,x_{R,1} - 0.62\,x_{O,5}$; the two features have correlation 0.997, so the two parties hold almost the same variable. R's own block is much better conditioned, $\lambda_{\min}(\mathbf{A}/n) = 0.010$. Neither party can see this collinearity in its own data.
- Block-normalized: the weakest direction lies inside R's block, $-0.66\,x_{R,1} + 0.69\,x_{R,2} - 0.27\,x_{R,3}$, and $\ell \approx \lambda_{\min}(\mathbf{A}/n) = 0.0021$: normalizing a block of strongly correlated features

leaves it close to degenerate.

Table 7 repeats the full preprocessing with one feature removed. Removing $x_{R,1}$ removes both weak directions: with block normalization ℓ rises to 0.013 and n_{off} falls from 691k to 102k at $\varepsilon = 1$ and from 186k to 27k at $\varepsilon = 4$. Removing $x_{O,5}$ instead helps only the standardized design.

Table 7: Protein: ℓ and n_{off} ($\delta = 10^{-9}$) with one feature removed before preprocessing; bounds recomputed from the reference split.

Features	Preprocessing	Δ	ℓ	n_{off}, $\varepsilon = 1$	n_{off}, $\varepsilon = 4$
all 9	standardize	87.3	0.00191	13.1 M	3.53 M
all 9	block unit norm	8.4	0.00201	691 k	186 k
without $x_{R,1}$	standardize	79.6	0.0130	1.69 M	455 k
without $x_{R,1}$	block unit norm	8.4	0.0131	102 k	27 k
without $x_{O,5}$	standardize	74.4	0.0100	2.03 M	547 k
without $x_{O,5}$	block unit norm	8.4	0.0021	647 k	174 k

Two diagnostics cost no budget. First, $\mathbf{A}$ is a principal submatrix of $\mathbf{G}$, so $\lambda_{\min}(\mathbf{G}) \le \lambda_{\min}(\mathbf{A})$ by Cauchy interlacing, and after linkage R obtains from its exact $\mathbf{A}$ the lower bound $n_{\text{off}} \ge n(g^\star + r_1)/\lambda_{\min}(\mathbf{A})$ before any release; O can bound ℓ the same way from its own block on the register. Second, a collinearity across the parties is invisible to both but visible on a public reference set that carries both schemas, where one exists, and screening features on it is a schema-level decision that can be made in Setup.

Levers. Table 8 lists what lowers n_{off}, at $\varepsilon = 1$ on the simulation population, where the evaluation settings give 41.8k.

Table 8: Levers on the ridge threshold, simulation population, $\varepsilon = 1$, $\delta = 10^{-9}$ (at the evaluation settings $n_{\text{off}} = 41.8$k).

Lever	n_{off}	Cost
Smaller Δ, larger ℓ: per-party transforms (Section 6.1); feature screening on public reference data	$\propto \Delta/\ell$	none in privacy; on protein a factor 6.8 from one feature
Exact $(1-\alpha_L)$ quantile for r_1 ($7.43\sigma_1$ instead of $11.53\sigma_1$)	32.1k (-23%)	none
$f_1 = 0.35$ instead of 0.1	31.7k (-24%)	σ_2 up 18% once the ridge is off
Both of the above	26.5k (-37%)	σ_2 up 18%
$\vartheta^\star = 2$ instead of 2.5	38.8k (-7%)	thinner validity margin (Remark 9.1)
A public lower bound on ℓ instead of stage 1	at best 14.6k (-65%)	validity rests on the reference data being representative; the gate still guards each release
Larger ε	$\propto 1/\sqrt{\rho_{\text{run}}} \approx 1/\varepsilon$	budget

Below n_{off} the ridge is not a tuning choice: it is what certifies the validity ratio $\vartheta^\star$ on which the expansion and the intervals rest (Remark 10.2). The levers lower the threshold; none of them makes the ridge unnecessary below it. A study can be planned by evaluating (9) at the intended ε, with ℓ estimated on public reference data or bounded through $\lambda_{\min}(\mathbf{A})$.

[**Check 16:** The threshold treats lambda-min of the noisy stage-1 Gram as n times ell and does not model its downward bias under noise; either derive a correction (it depends on how many eigenvalues of Sigma lie near ell) or keep the empirical factor 1.2.]

[**Check 17:** Decide whether the exact-quantile r-1 and f-1 near 0.35 become the recommended settings (they lower the threshold by 37 percent together); coverage would need to be re-checked.]

[**Check 18:** Decide whether feature screening on public reference data becomes a Setup step, and what reference data it requires.]

9.6 Status of the statistical guarantees

The companion proofs (Part B) establish: the operator-norm radius r_1 for the one-sided event (exact); the independence and closed forms of the two-stage rule; the bound on $\|\boldsymbol{\beta}\|$ for the projection coefficient with no model assumed; the Wald remainder exactly, with the plug-in variance to first order and normality of the sampling term asymptotically; and the bootstrap RSS correction to first order, without explicit constants. A remedy for heteroskedasticity when clipping binds is open.

10 Mechanism bias and the optional correction

Throughout this section σ denotes the stage-2 slot s.d. σ_2, and (λ, Ψ) is fixed independently of the stage-2 noise (Proposition 9.1).

10.1 Noise model

Write the stage-2 statistics as $\hat{\mathbf{G}}_2 = \mathbf{G} + \mathbf{E}$ and $\hat{\mathbf{c}}_2 = \mathbf{c} + \mathbf{f}$, with

$$\mathbf{E} = \begin{bmatrix} \mathbf{0} & \mathbf{E}_C \\ \mathbf{E}_C^\top & \mathbf{E}_B \end{bmatrix}, \qquad \mathbf{f} = \begin{bmatrix} \mathbf{f}_R \\ \mathbf{f}_O \end{bmatrix}. \tag{10}$$

The top-left block of $\mathbf{E}$ is identically zero, since $\mathbf{A}$ is exact. Each independent entry on and above the diagonal of the O-dependent blocks is drawn with variance σ^2 and mirrored, so every entry, the diagonal included, has variance σ^2. With $\chi_{ij} = 0$ iff i, j both lie in R's block and $\chi_{ij} = 1$ otherwise,

$$\mathbb{E}[E_{ij}E_{kl}] = \sigma^2 \chi_{ij}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{ik}\delta_{il}\big). \tag{11}$$

The last term removes the diagonal double-count of the GOE convention. $\mathbf{E}$ and $\mathbf{f}$ occupy disjoint entries of the release, are independent, and are symmetric about zero. (The lattice noise induces slot-domain noise with these first and second moments; the CKKS error is flooded and of lower order.)

Condition on the realized sample, so $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$ is fixed, and set $\mathbf{P} = \mathbf{G}^{-1}$,

$$\vartheta_0 = \frac{\lambda_{\min}(\mathbf{G})}{2\sigma\sqrt{p}}, \qquad \mathcal{E}_\eta = \big\{\|\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}\|_{\mathrm{op}} \le \eta\big\} \quad (0 < \eta < 1). \tag{12}$$

$\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$. Define

$$M(\mathbf{P}) = \big(\mathbf{P}^2 + \mathrm{tr}(\mathbf{P})\mathbf{P} - \mathbf{P}\,\mathrm{diag}(\mathbf{P})\big)\Pi_O + \big(\mathbf{P}\Pi_O\mathbf{P} + \mathrm{tr}(\Pi_O\mathbf{P})\mathbf{P}\big)\Pi_R, \tag{13}$$

where $\mathrm{diag}(\mathbf{P})$ is the diagonal matrix formed from the diagonal of $\mathbf{P}$.

Remark 10.1 (The unconditional mean does not exist)**.** With Gaussian $\mathbf{E}$, $(\mathbf{G}+\mathbf{E})^{-1}(\mathbf{c}+\mathbf{f})$ has no finite mean: already for $p=1$, $\mathbb{E}[1/(a+e)]$ diverges. Statements about its bias must therefore be conditional, or about a truncated estimator. We condition on $\mathcal{E}_\eta$, whose complement has exponentially small probability.

10.2 Result

Theorem 10.1 (Leading-order conditional bias)**.** *Fix $\eta \in (0,1)$ with $\eta\vartheta_0 > 1$. Then, for $\tilde{\boldsymbol{\beta}} = (\mathbf{G}+\mathbf{E})^{-1}(\mathbf{c}+\mathbf{f})$,*

$$\mathbb{E}\big[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta\big] - \hat{\boldsymbol{\beta}} = \sigma^2 M(\mathbf{P})\,\hat{\boldsymbol{\beta}} + O(\sigma^4) + O\big(e^{-p(\eta\vartheta_0-1)^2/2}\big),$$

where the constants depend on $\|\mathbf{P}\|_{\mathrm{op}}$, $\|\hat{\boldsymbol{\beta}}\|$, p and η. The moment-vector noise $\mathbf{f}$ contributes nothing at this order; it is variance-only. The term $-\mathbf{P}\,\mathrm{diag}(\mathbf{P})\Pi_O$ is the single-draw correction to the GOE contraction.

The proof is given in Appendix A. Setting $\Pi_O \to \mathbf{I}$ and $\Pi_R \to \mathbf{0}$ recovers the homogeneous-Gram bias, the correct check when every block is noised.

Remark 10.2 (When the expansion is valid)**.** $\mathbf{PE}$ is similar to $\mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}$, so the Neumann series for $(\mathbf{I}+\mathbf{PE})^{-1}$ converges iff $-\mathbf{G} \prec \mathbf{E} \prec \mathbf{G}$; on $\mathcal{E}_\eta$ it converges geometrically with ratio η. $\|\mathbf{E}\|_{\mathrm{op}}$ concentrates at or below $2\sigma\sqrt{p}$ with sub-Gaussian tails, so $\Pr(\mathcal{E}_\eta^c) \le \exp(-p(\eta\vartheta_0-1)^2)$ for $\eta\vartheta_0 > 1$. Since $\lambda_{\min}(\mathbf{G})$ grows linearly in n while σ does not depend on n, the condition holds with high probability once n is large enough; below that, a ridge is required.

10.3 Schur-complement reading

Let $\mathbf{S} = \mathbf{B} - \mathbf{C}^\top\mathbf{A}^{-1}\mathbf{C}$ be the Schur complement of $\mathbf{A}$ in $\mathbf{G}$. Block inversion gives $\Pi_O\mathbf{P}\Pi_O = \mathbf{S}^{-1}$, hence

$$\mathrm{tr}(\Pi_O\mathbf{P}) = \mathrm{tr}(\mathbf{S}^{-1}), \qquad \mathrm{tr}(\mathbf{P}) = \mathrm{tr}(\mathbf{A}^{-1}) + \mathrm{tr}\big(\mathbf{A}^{-1}\mathbf{C}\mathbf{S}^{-1}\mathbf{C}^\top\mathbf{A}^{-1}\big) + \mathrm{tr}(\mathbf{S}^{-1}). \tag{14}$$

The two trace weights governing the leading bias are $\mathrm{tr}(\mathbf{G}^{-1})$ for the O-coefficients and $\mathrm{tr}(\mathbf{S}^{-1})$ for the R-coefficients. When O's features carry little signal beyond R's, $\mathbf{S}$ is near-singular and the private estimator is worst-biased in R's own coefficients.

10.4 Ridge

Let $\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi$, $\mathbf{P}_\lambda = \mathbf{G}_\lambda^{-1}$, and let ϑ_λ and $\mathcal{E}_{\eta,\lambda}$ be as in (12) with $\mathbf{G}$ replaced by $\mathbf{G}_\lambda$.

Theorem 10.2 (Ridge mechanism bias)**.** *Suppose (λ,Ψ) is independent of $(\mathbf{E},\mathbf{f})$ and $\eta\vartheta_\lambda > 1$. Then*

$$\mathbb{E}\big[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}\big] - \boldsymbol{\theta}_\lambda = \sigma^2 M(\mathbf{P}_\lambda)\,\boldsymbol{\theta}_\lambda + O(\sigma^4) + O\big(e^{-p(\eta\vartheta_\lambda-1)^2/2}\big),$$

with the moment-vector noise $\mathbf{f}$ again variance-only.

The proof is that of Theorem 10.1 under $\mathbf{G} \mapsto \mathbf{G}_\lambda$, which leaves $\mathbf{E}$, its covariance (11) and $\mathbf{f}$ untouched.

Proposition 10.1 (Total bias against least squares: shrinkage split)**.** *If the estimand is* $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$*, the total bias of the private ridge splits as*

$$\mathbb{E}[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}] - \hat{\boldsymbol{\beta}} = \underbrace{\boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}}}_{\textit{deterministic}} + \underbrace{\mathbb{E}[\tilde{\boldsymbol{\theta}} \mid \mathcal{E}_{\eta,\lambda}] - \boldsymbol{\theta}_\lambda}_{\textit{mechanism}}, \qquad \boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}} = -\lambda \mathbf{P}_\lambda \Psi \hat{\boldsymbol{\beta}}.$$

The deterministic term grows in λ*; the mechanism term decays like* σ^2/λ*.*

For $\Psi = \mathbf{I}$, $\lambda_{\min}(\mathbf{G}_\lambda) = \lambda_{\min}(\mathbf{G}) + \lambda$ grows without bound. For $\Psi = \Pi_O$, $\lambda_{\min}(\mathbf{G} + \lambda\Pi_O) \to \lambda_{\min}(\mathbf{A})$ as $\lambda \to \infty$, so the O-block ridge cannot validate the expansion when $\lambda_{\min}(\mathbf{A})$ is too small, which is why the stage-1 rule falls back to $\Psi = \mathbf{I}$.

10.5 Why the ridge must not depend on the estimation noise

Earlier drafts chose (λ, Ψ) from the same noisy Gram matrix used for estimation. That choice is post-processing and costs no privacy, but it makes λ a function of $\mathbf{E}$ at first order, so Theorem 10.2 no longer applies. In simulation, with λ fixed independently the correction below removed 87–98% of the mechanism bias, whereas under the adaptive rule it removed essentially none: the residual after correction was 1.0–1.6 times the uncorrected bias. The two-stage rule of Section 9.2 keeps the adaptivity and restores independence.

10.6 Optional zero-budget correction

Since σ, d_R, d_O and Π_O are public and $\hat{\mathbf{G}}_2$ is already private, the bias is estimable by post-processing:

$$\tilde{\boldsymbol{\theta}}_{\mathrm{bc}} = \tilde{\boldsymbol{\theta}} - \sigma^2 M(\tilde{\mathbf{P}})\, \tilde{\boldsymbol{\theta}}. \tag{15}$$

Substituting $\hat{\mathbf{G}}_2$ for $\mathbf{G}$ inside a term already of order σ^2 injects only $O(\sigma^3)$ fluctuation and $O(\sigma^4)$ bias. One must not additionally invert the deterministic shrinkage, which would re-amplify precisely the directions the ridge protects. In practice the bias removed is small: in a single release the mechanism bias is at most 0.75% of the mean squared error (Section 13.6). The correction is therefore an option of the protocol, off in our evaluation.

[**Check 19:** Decide whether this section stays in the main text, given that the correction is off in the evaluation, or moves to an appendix.]

11 Prediction after training

[**Check 20:** This section is carried over from the earlier draft and adapted to threshold keys. Split inference is not part of the v3.1 specification: the sensitivity, noise and ledger charge of a prediction release are still to be defined.]

After a run, R holds $\tilde{\boldsymbol{\theta}} = (\tilde{\boldsymbol{\theta}}_R; \tilde{\boldsymbol{\theta}}_O)$. For a new individual i, $\hat{y}_i = \dot{\mathbf{x}}_i^\top \tilde{\boldsymbol{\theta}}_R + \mathbf{x}_{O,i}^\top \tilde{\boldsymbol{\theta}}_O$; R can evaluate the first term but has no access to $\mathbf{x}_{O,i}$.

Approach 1: split inference. R encrypts $\tilde{\boldsymbol{\theta}}_O$ under the joint key; after linkage on the test identifiers, O computes $\mathsf{Enc}(p_{O,i}) = \sum_k X_{O,ik} \mathsf{Enc}\left(\tilde{\theta}_{O,k}\right)$ and releases it with Algorithm 5. Threshold decryption fits this use well. Every prediction needs O's share, so O adds noise to each batch and charges it to the ledger. Re-randomization prevents the attack in which R

solves the linear system in c_1 for $\mathbf{x}_{O,i}$ exactly. Two leaks remain and must be priced: each decrypted $p_{O,i}$ is a known-direction projection of $\mathbf{x}_{O,i}$; and across k repeated queries or $T \geq d_O$ model versions, averaging or solving the projections reconstructs $\mathbf{x}_{O,i}$ unless every query and retraining is charged.

Approach 2: distillation to an R-only model. The student coefficient that regresses the joint model's fitted values on R's features is $\hat{\boldsymbol{\beta}}_R^{\mathrm{st}} = \mathbf{A}^{-1}\mathbf{c}_R = \boldsymbol{\beta}_R + \mathbf{A}^{-1}\mathbf{C}\boldsymbol{\beta}_O$. R computes $\mathbf{A}^{-1}\hat{\mathbf{c}}_R$ from its exact $\mathbf{A}$ and the released $\hat{\mathbf{c}}_R$, which is post-processing and needs no interaction at test time. Its loss against the joint fitted values is $\boldsymbol{\beta}_O^\top \mathbf{S} \boldsymbol{\beta}_O$, the quadratic form of $\boldsymbol{\beta}_O$ in the Schur complement: when it is small, distillation suffices; when large, O's features carry signal that R's cannot proxy.

12 Parameters

The protocol is defined for a parameter vector $\mathcal{P}$. Table 9 lists every parameter: its role, the constraint it must satisfy, and the value used in this paper's evaluation. The last column is an *evaluation setting*, one admissible choice among many; it is not part of the protocol.

The constraints are of three kinds. Some are hypotheses that $\mathcal{P}$ itself must meet, such as the ranges of the shares and the fact that the bounds and the preprocessing constants are public and fixed in advance; the parties check them in Setup. Some involve quantities that CALIBRATE derives: the existence of κ, $\delta_{\mathrm{DP}} > 0$, a scale that floods, the size of q_0 and the security bound on the chain. CALIBRATE checks these and refuses the run otherwise (Section 7.3). The rest, such as f_1, $\vartheta^\star$ and κ_Ψ, affect only utility or cost and need only lie in their ranges. We call $\mathcal{P}$ *admissible* if all constraints of Table 9 hold. Theorem 7.2, Proposition 7.2, the input-privacy and correctness statements of Section 8 and the statistical results of Section 9 hold for every admissible $\mathcal{P}$, with the constants they state; none of them uses a particular value.

Table 9: Parameters of the protocol. "Derived" marks quantities that CALIBRATE computes from the others. The last column is the evaluation setting, not part of the protocol.

Parameter	Role	Constraint	Evaluation setting
Security			
N	CKKS ring dimension; $N_s = N/2$ slots	power of two; with the chain, $\log_2(QP) \leq L_{\max}(N, \lambda_{\mathrm{sec}})$	2^{14}
λ_{sec}	computational security level	sets $L_{\max}$ [1]: 438 for $N = 2^{14}$ at 128 bits, ternary secrets	128 bits (classical)
chain q_0, q_1, q_2; P; d_{num}	last-level modulus; primes removed by the two rescales; special modulus and digits of key switching	$\log_2(QP) \leq L_{\max}$; $q_0 > 2(\Omega_f S_{\max} + 2^\kappa B_{\mathrm{tot,max}} + t_\eta \sigma_{\mathrm{c,max}})$ (checked by CALIBRATE)	q_0: primes of 60 and 59 bits; q_1, q_2: 59 bits; P: two 60-bit primes; $d_{\mathrm{num}} = 2$; $\log_2(QP) = 357$
secret and error distributions	RLWE assumptions of Theorem 7.2 and G1	uniform ternary secrets; discrete Gaussian errors	$\sigma_e = 3.19$ (bounds use 3.2)
δ_B	failure probability of $B_{\mathrm{tot},s}$	$\delta_B \in (0,1)$; $n_{\mathrm{rel}}(1+e^{\varepsilon_{\mathrm{run}}})(\delta_B + 2^{-97}) < \zeta\delta_{\mathrm{run}}$, and $n_{\mathrm{rel}}(\delta_B + 2^{-97})$ below the run's ledger share	2^{-50}

continued on the next page

Table 9, continued

Parameter	Role	Constraint	Evaluation setting
$\kappa_{\min}$	floor on the flooding parameter (statistical security)	integer ≥ 1	40
κ	flooding parameter	derived by (8); if fixed instead, $\delta_{\mathrm{DP}} > 0$	derived (Table 4)
t_η	noise margin in q_0, in units of $\sigma_{c,s}$	$t_\eta > 0$; correctness failure $2Ne^{-t_\eta^2/2}$ per release	10
Privacy of a run			
$(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$	budget of a run	$\varepsilon_{\mathrm{run}} > 0$, $\delta_{\mathrm{run}} \in (0,1)$; admitted by the ledger	$\varepsilon \in \{0.5, 1, 2, 4, 8\}$, $\delta = 10^{-9}$
f_1	share of ρ_{run} for stage 1	$f_1 \in (0,1)$, $f_1 + f_y < 1$	0.1
f_y	share of a separate $\mathbf{y}^\top\mathbf{y}$ release (separate mode only)	$f_y \in (0, 1 - f_1)$; no protocol default	0.1 in the implementation; every evaluation run is in joint mode
ζ	largest share of δ_{run} the slack may take	$\zeta \in (0,1)$	0.1
mode, n_{rel}	$\mathbf{y}^\top\mathbf{y}$ in stage 2 or apart; number of releases	derived: joint ($n_{\mathrm{rel}} = 2$) iff $\mathbf{y}^\top\mathbf{y}$ is free (Remark 7.1)	derived
Ledger of a base register			
$(\varepsilon_L, \delta_L)$	cap over all runs on the register	$\varepsilon_L \geq \varepsilon_{\mathrm{run}}$, $\delta_L \geq \delta_{\mathrm{run}}$ for every run; public	one run per register: cap $= (\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$; example $(10, 10^{-6})$ in Table 4
ζ_L	share of δ_L reserved for slack	$\zeta_L \in (0,1)$	0.1
admission rule	runs fixed in advance, or chosen adaptively	the filter rule unless every run is fixed in advance (Section 7.4)	fixed (one run)
OPRF quota	membership look-ups per researcher and project	policy	none
Statistics			
$\vartheta^\star$	validity target of the ridge	$\vartheta^\star > 1$; the refusal bound needs $\vartheta^\star > 2$; coverage is asymptotic in $\vartheta^\star$ (Remark 9.1)	2.5
κ_Ψ	when to shrink only O's block	$\kappa_\Psi \geq 0$	2
$\alpha = \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U$	error level and its split	each part > 0, $\alpha < 1$	$\alpha = 0.05$, $\alpha_L = \alpha_U = 0.0025$
η_B, B	bootstrap truncation and number of draws	$\eta_B \in (0,1)$; Monte Carlo error of relative order $B^{-1/2}$	1/2; 2000
B_β^{pub}	public a-priori bound on $\lVert\boldsymbol{\beta}\rVert$	> 0, or ∞	∞ unless stated
bias correction	optional correction (15)	on or off	off
Data			
B_R, B_O, B_y	clipping bounds on the preprocessed data	public and fixed before the data are used (reference data or domain knowledge); $B_R \geq 1$ covers the intercept	99.5% quantiles on a public reference split; simulation population $(5.59, 6.24, 2.89)$; block normalization: $B_R^2 = 2$, $B_O = 1$

continued on the next page

Table 9, continued

Parameter	Role	Constraint	Evaluation setting
preprocessing	per-party transforms (Section 6.1)	fixed functions of one record and public constants (Proposition 6.1)	standardize; optionally winsorize at $\|z\| \le 2$ or block unit norm
Φ, $\bar{n}_O$	study scope; padded register size	Φ on public attributes only; $\bar{n}_O \ge \|D_\Phi\|$, public	whole register; $\bar{n}_O$ from 4,096 to 65,536
$n_R^{\max}$	certified cohort cap	$n_R \le n_R^{\max}$, public	cohort size plus a margin
Labeled PSI			
N_B, t	BFV ring dimension; plaintext modulus, which is the item length	t prime, $t \equiv 1 \bmod 2N_B$; $t \ge n_R^{\max}\bar{n}_O h/(N_{\text{bins}}\delta_{\text{psi}})$ for false matches (Appendix B)	2^{14}; $t \approx 2^{32}$, which does not meet the constraint (Appendix B)
h, N_{bins}, d_{cap}, δ_{psi}	cuckoo hashing, bins, largest partition degree, failure probability	sizes are public functions of these; failures are correctness events, not charged to δ	3; 2^{14}; 64; 2^{-40}
κ_ψ, B_ψ, $\delta_{B,\psi}$, q'	flooding of the PSI replies (SanitizeBFV)	$\sigma_\psi \ge \max\{2^{\kappa_\psi} B_\psi/\sqrt{2\pi}, 2^{20}\}$; $q'/(2t) > t_\eta\sigma_\psi + B_\psi$; B_ψ a worst-case bound (Section 8)	40; 2^{21} (evaluation noise measured, not bounded); 0; two 60-bit limbs
Encoding			
Ω_U	scale of R's vectors	exactly representable; $\Omega_U \ge \Omega_f$ keeps B_{tot} independent of $\bar{n}_O$	2^{68}
Ω_u	scale of the one-hot masks	exactly representable	q_1
Ω_V, Ω_f	scale of O's data; final scale	derived: smallest $\Omega_f = \Omega_U\Omega_V\Omega_u/(q_1 q_2)$ that floods, tracked exactly	derived (Table 4)

Derived quantities. Table 10 follows Calibrate through one run at the evaluation settings.

The chain in the evaluation. The implementation uses $N = 2^{14}$, a ciphertext modulus of four primes of 60, 59, 59 and 59 bits (q_0 is the product of the first two, about 2^{119}; q_1 and q_2 are the others), two 60-bit special primes and decomposition number 2, for $\log_2(QP) = 357$, below the 128-bit limit of 438 for $N = 2^{14}$. A 64-bit build of OpenFHE keeps every prime below 2^{60}. Honest inputs need q_0 of 80–92 bits over $\bar{n}_O \in [4{,}096, 10^6]$ and $\varepsilon \le 4$, and 98 bits at $\varepsilon = 8$ (Table 4); the specification's "70–73 bits" is too small. $N = 2^{15}$ brings no gain: it makes the aggregation at 10^6 rows about 40% slower.

Table 10: Example: the derived quantities at the evaluation settings for the simulation population ($d_R = 4$, $d_O = 6$, $p = 11$), $\varepsilon_{\text{run}} = 1$, $\delta_{\text{run}} = 10^{-9}$, a single-run ledger and $\bar{n}_O = 4{,}096$, as computed by the implementation.

Quantity	How	Value
Δ, Δ_G	DeltaRep (Proposition 7.1)	86.15, 77.16
mode, n_{rel}	$\mathbf{y}^\top\mathbf{y}$ free (Remark 7.1)	joint, 2
κ	(8); $\kappa_{\text{run}} = 37$, so the floor binds	40
γ; slack $(1 + e^{\varepsilon})\gamma$	step 2	1.82×10^{-12}; 0.68% of δ_{run}
δ_{DP}; ρ_{run}	step 3, tight conversion	9.93×10^{-10}; 0.01497
ρ_1, ρ_2	split by f_1	0.00150, 0.01347
σ_1, σ_2	(7)	1410.3, 524.9
Ω_V; Ω_f	step 4	$2^{53.75}$; $2^{62.75}$
$B_{\text{tot},1}$, $B_{\text{tot},2}$	Theorem 7.1, OpenFHE's gadget	$2^{26.43}$, $2^{26.59}$
$\Delta_{x,1}$, $\Delta_{x,2}$	(6)	$2^{27.46}$, $2^{27.57}$
$\sigma_{c,1}$, $\sigma_{c,2}$	$\lceil \widehat{\Delta}^2_{\text{enc},s}/(2\rho_s) \rceil^{1/2}$	$2^{66.71}$, $2^{65.29}$
q_0 needed	step 5	$2^{81.0}$ ($2^{89.0}$ at $\bar{n}_O = 10^6$); available 2^{119}
$\log_2(QP)$	chain	$357 \le 438$

13 Empirical evaluation

Two kinds of results are reported. Sections 13.2 and 13.3 use protocol v3.1 as specified: its calibration and a simulation of its release and solve steps. Sections 13.5–13.9 use an *earlier single-release mechanism*: one Gaussian release per run with the analytic-Gaussian σ at $\delta \in \{10^{-5}, 10^{-6}\}$, a ridge $\lambda = \max\{0, 2\vartheta^\star\sigma\sqrt{p} - n\ell\}$ with $\vartheta^\star = 2$ set from a committed eigenvalue bound ℓ, and the bias correction applied. At the same (ε, δ) the two releases of v3.1 make σ_2 1.11–1.16 times larger (Table 12); at $\delta = 10^{-9}$ it is 1.45–1.63 times the σ used in the simulations. Since the records needed for coefficient accuracy scale like σ^2, the crossings reported below should move to roughly 2–2.7 times larger n under v3.1, and the stage-1 ridge stays on to larger n than the committed-bound rule (Table 13). The scripts that produce every number are in the `experiments/` directory of the project repository.

[**Check 21:** Rerun Sections 13.4 to 13.8 under v3.1: two releases with the tight conversion, the ridge from stage 1, delta = 1e-9 (or the chosen default), the bias correction off, and the gate.]

13.1 The cryptographic layer is exact

A prototype of the linkage and aggregation steps was run end to end over sockets, on six cohorts ($n = 200$ to 64,000 matched records out of $n_O = 2{,}000$ to 80,000) with a plaintext backend, and on one cohort ($n = 400$ out of $n_O = 4{,}000$) with OpenFHE 1.5.1 (BFV, $N = 16{,}384$, for the linkage; CKKS with scale 2^{50} for the aggregation). Labeled PSI recovered exactly the true match set in every run, with no misaligned rows (count, set and row-by-row alignment checked). With the noise switched off, the regression agreed with its plaintext target to 2.9×10^{-10} with the plaintext backend and 1.7×10^{-10} with OpenFHE. The OpenFHE run took about 13 s end to end. The prototype hashes identifiers with SHA-256 instead of an OPRF, uses a single CKKS key, and has no release step.

Full implementation of v3.1. The `implementation/` directory of the repository implements every phase of v3.1 on OpenFHE 1.5.1, with R and O as separate processes and a transcript of every message: RFC 9497 OPRF (ristretto255), labeled PSI under R's BFV key with flooded replies, threshold key generation, the seeded upload and key conversion, the aggregation circuit, and the release with exact discrete-Gaussian noise inside O's share. A small C++ layer provides the polynomial-level access that the release needs. On the toy example and on a 4,096-row synthetic register it recovers exactly the true match set and alignment; with the noise switched off by a test-only flag the decrypted statistics match their plaintext values to 10^{-11} relative; with the noise on, the measured noise s.d. is $0.996\,\sigma_2$ and $0.976\,\sigma_1$; the randomness-dependent encryption error stays about 8 bits below the worst-case B_{tot}; and R never receives c_0' while O never receives anything that could serve as a decryption share of R. Its deviations from the specification are listed in `implementation/DEVIATIONS.md`.

13.2 Calibration under v3.1

Table 11 gives σ_2/Δ for two releases with $f_1 = 0.1$ and $\mathbf{y}^\top\mathbf{y}$ free; Table 12 compares it with the single analytic-Gaussian release the earlier experiments simulated. The planning formula of (7) is conservative by 6–33% over $\varepsilon \in [0.1, 8]$ and $\delta \in [10^{-9}, 10^{-6}]$.

Table 11: Stage-2 noise σ_2/Δ under v3.1 (two releases, $f_1 = 0.1$, tight conversion).

$\delta \setminus \varepsilon$	0.1	0.25	0.5	1	2	4	8
10^{-6}	41.6	17.6	9.15	4.78	2.51	1.34	0.727
10^{-8}	51.6	21.4	11.0	5.68	2.95	1.55	0.825
10^{-9}	56.0	23.1	11.9	6.09	3.15	1.64	0.871

Table 12: σ_2 of v3.1 divided by the single-release analytic-Gaussian σ at the same (ε, δ).

$\delta \setminus \varepsilon$	0.1	0.25	0.5	1	2	4	8
10^{-5}	1.16	1.16	1.15	1.14	1.14	1.13	1.12
10^{-6}	1.15	1.14	1.14	1.13	1.13	1.12	1.11

Table 13 works the calibration through for the simulation population of Section 13.5 at $\delta = 10^{-9}$. Learning the ridge from stage 1 costs data: at $\varepsilon = 1$ the ridge stays on until about 51k matched records, against 16k for a single release with the eigenvalue bound known in advance.

Table 13: Worked calibration, simulation population ($B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $p = 11$, $\Delta = 86.15$, $\Delta_G = 77.16$, population eigenvalue bound $\ell = 0.494$), $\delta = 10^{-9}$. "Ridge off from": the n above which the stage-1 rule gives $\lambda = 0$ (typically), and for comparison a single release with ℓ known.

ε	ρ_{run}	σ_1	σ_2	r_1	λ at $n = 10$k	ridge off from	one release, ℓ known
0.5	0.0040	2744	1021	31637	43633	98k	31k
1	0.0150	1410	525	16256	20018	51k	16k
2	0.0561	728	271	8396	7951	26k	8.2k
4	0.2063	380	141	4379	1784	14k	4.3k
8	0.7328	202	75	2324	0	7.2k	2.3k

13.3 Interval coverage under v3.1

Table 14 simulates the release and solve steps of Algorithm 6 (2,000 replications per scenario, $p = 11$), in scenarios chosen to exercise each branch of the ridge rule. The Wald interval hits its target with or without the ridge. The bias-aware interval for $\boldsymbol{\beta}$ is conservative, because it uses the worst-case bias and a cautious variance, and it is unbounded when the ridge is on and no public bound on $\|\boldsymbol{\beta}\|$ is given. For contrast, a naive Wald interval around a ridge estimate, read as an interval for $\boldsymbol{\beta}$, covered only 42–81% in the earlier simulations when no eigenvalue bound was available.

Table 14: Coverage of both intervals over the 11 coefficients (simulation of the release and solve steps; noise levels set directly).

Scenario	n	Ridge	Wald for $\boldsymbol{\theta}_\lambda$ (target 0.955)	Bias-aware for $\boldsymbol{\beta}$ (target ≥ 0.95)
Ridge off	4,000	$\lambda = 0$	0.946–0.958	0.945–0.968
Ridge on, whole matrix	1,500	$\Psi = \mathbf{I}$, median $\lambda = 477$	0.948–0.960	1.000 with a public bound on $\|\boldsymbol{\beta}\|$; unbounded without
Ridge on, O's block only	3,000	$\Psi = \Pi_O$, median $\lambda = 439$	0.944–0.962	unbounded without a public bound
Small n, public bound	200	$\Psi = \mathbf{I}$, median $\lambda = 1{,}021$	0.946–0.962	1.000

[**Check 22:** The coverage table uses noise levels set directly (sigma-1 = 2.5 sigma-2). Repeat it with the calibrated sigma-1 and sigma-2 of a real budget, and on real data.]

13.4 End-to-end runs of the implementation

These runs use the full implementation: OPRF and labeled PSI, threshold CKKS upload and aggregation, two releases with exact discrete-Gaussian noise inside O's share, and the solver, with $\delta = 10^{-9}$ and about 40–45 s per run on one core per party. They were made with a fixed flooding parameter, $\kappa = 40$ at $\varepsilon \in \{1, 4\}$ and $\kappa = 48$ at $\varepsilon = 8$, before the rule (8), which gives 40, 41 and 46 at the evaluation settings. The difference changes Ω_f but σ_1 and σ_2 by less than 0.2%.

Real data. The protein dataset (45,730 rows, 9 features) is split into a public reference set of 10% (standardization and transform constants, 99.5% bounds), a test set of 20%, and O's register (32,011 rows, padded to 32,768). R holds 4 features and O the other 5 and y. R's cohort is 19,206 register members plus 1,000 people O does not hold. Table 15 gives five encrypted runs per setting. Every median lies inside the 95% range of a plaintext simulation of the same mechanism, so the encryption does not change what the mechanism produces. Pooled over all 30 runs, the Wald interval covers $\boldsymbol{\theta}_\lambda$ for 286 of 300 coefficients (target 0.955).

Block normalization costs a little in the non-private fit (0.745 against 0.728) and gains much more in the private one: at $\varepsilon = 4$ its private error, 0.759, is within 4% of the best non-private model. The ridge is on in every setting, so the coefficients remain shrunk (63–82% of the

Table 15: Protein, 19,206 matched records: test MSE of the private estimate (median of 5 encrypted runs), the 95% range of a plaintext simulation of the same mechanism (2,000 draws), non-private least squares under the same preprocessing, their ratio, and the median ridge. Predicting zero gives 1.010.

Preprocessing	Δ	ε	Encrypted	Simulated range	Non-private	Ratio	λ
standardize	87.3	1	0.883	0.871–0.896	0.728	1.21	29,678
standardize	87.3	4	0.815	0.807–0.821	0.728	1.12	7,321
winsorize $\lvert z\rvert \le 2$	40.5	1	0.850	0.838–0.855	0.726	1.17	12,552
winsorize $\lvert z\rvert \le 2$	40.5	4	0.788	0.784–0.795	0.726	1.09	3,088
block unit norm	8.4	1	0.793	0.786–0.795	0.745	1.06	1,385
block unit norm	8.4	4	0.759	0.757–0.760	0.745	1.02	338

Table 16: Median RMSE of the standardized slopes against the true coefficients, $R^2 = 0.5$, earlier mechanism, with the ratio to the non-private floor in brackets.

n	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	OLS floor
10^3	0.231 (9.9)	0.213 (9.1)	0.181 (7.7)	0.138 (5.9)	0.078 (3.3)	0.0234
10^4	0.092 (12.3)	0.044 (5.8)	0.025 (3.4)	0.014 (1.9)	0.010 (1.3)	0.0075
10^5	0.0087 (3.4)	0.0051 (2.0)	0.0034 (1.4)	0.0029 (1.2)	0.0026 (1.0)	0.0025

way from least squares to zero with block normalization); prediction recovers faster than the coefficients do. Section 9.5 predicts these ridges to within 1–19% and traces them to a near-collinearity that keeps the ridge on until 186k–13M matched records.

Synthetic data with the ridge off. With the simulation population of Section 13.5, 20,000 matched records in a 24,000-row register and fixed data, 20 encrypted runs per setting gave $\lambda = 0$ every time. The estimate was within 4.0% ($\varepsilon = 4$, $\kappa = 40$) and 2.5% ($\varepsilon = 8$, $\kappa = 48$) of least squares (median relative distance), and the Wald interval covered $\boldsymbol{\theta}_\lambda$ for 0.950 (209/220) and 0.927 (204/220) of the coefficients, with standardized errors of mean +0.04 and +0.03 and s.d. 1.02 and 1.05.

[**Check 23:** Coverage at epsilon = 8 (0.927) and for some protein settings (46 of 50) is slightly below 0.955. Repeat with about 100 runs per setting before drawing conclusions.]

13.5 Coefficient accuracy against intersection size (earlier mechanism)

Setup. The population has $d_R = 4$ and $d_O = 6$ features on realistic raw scales (a score, a binary indicator, lognormal income and assets, a count and an age), Toeplitz-correlated with coefficient 0.3. They are standardized with public population constants and clipped to committed 99.5% bounds ($B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$). The generating slopes give $R^2 \in \{0.3, 0.5, 0.9\}$. The replace-one sensitivity is $\Delta = 86.2$; with $\delta = 10^{-5}$ the analytic Gaussian gives $\sigma = 606, 322, 172, 93, 52$ for $\varepsilon = 0.5, 1, 2, 4, 8$. The ridge uses $\vartheta^\star = 2$, $p = 11$ and the committed bound $\ell = 0.8\,\lambda_{\min}(\mathbf{G}/n) = 0.494$ of the population. For each $n \in [10^2, 10^5]$ we draw 200 cohorts and an independent release per ε, and report the bias-corrected estimate.

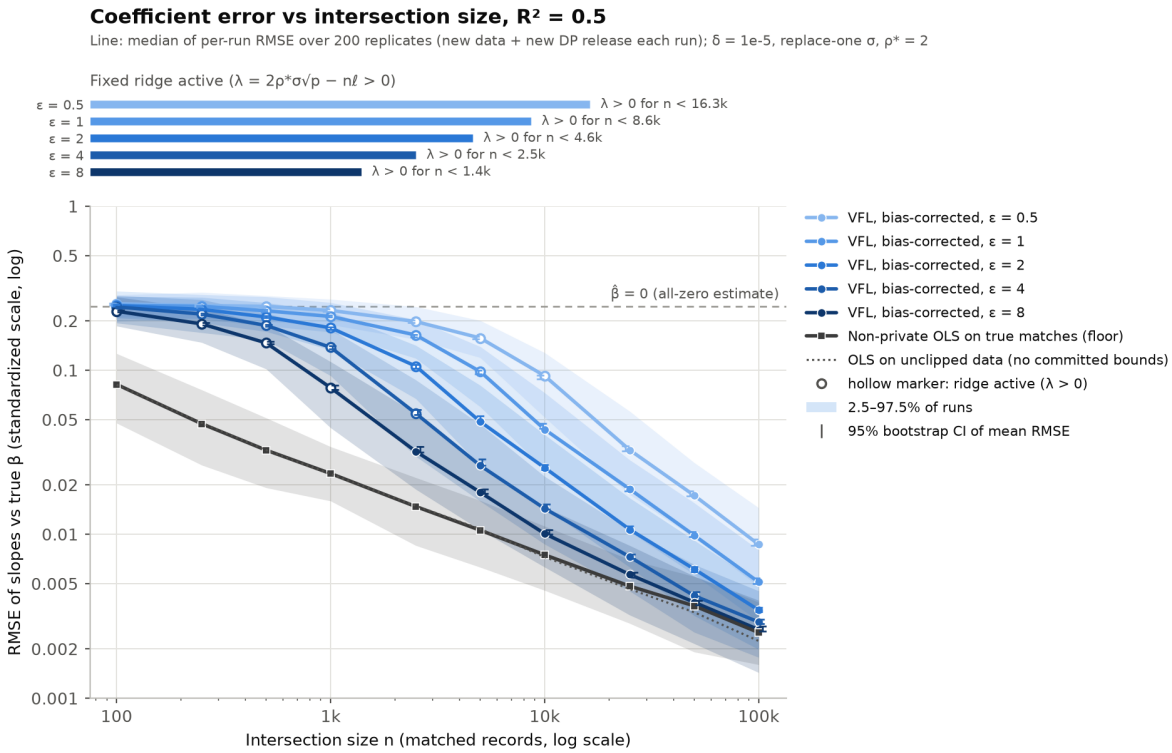


Figure 2: RMSE of the bias-corrected slopes against the true coefficients (standardized scale), against the intersection size n, at $R^2 = 0.5$, under the earlier single-release mechanism. Lines: median over 200 runs, each with a new cohort and a new release. Bands: 2.5–97.5% of runs. Error bars: bootstrap 95% CI of the mean. Hollow markers: the ridge is active. Grey: non-private least squares on the true matches.

Results. Figure 2 and Table 16 show three regimes. While the ridge is active ($n < 16.3$k, 8.6k, 4.6k, 2.5k and 1.4k for $\varepsilon = 0.5, \ldots, 8$), the estimate is shrunk towards zero. Once the ridge switches off, the DP error falls like $1/n$, faster than the sampling error's $1/\sqrt{n}$, and merges into the floor. The error reaches twice the floor at $n \approx 3.2$k ($\varepsilon = 8$), 9.4k ($\varepsilon = 4$) and 34k ($\varepsilon = 2$); at $\varepsilon = 1$ the ratio is still 2.04 at $n = 10^5$ (about 105k by extrapolation). A stronger signal lowers the floor but not the DP error, so the ratio is largest at $R^2 = 0.9$, where the same crossings are at 13k, 30k and 66k.

Coefficients on different scales. Raw-scale errors cannot be compared across coefficients: on this population they span 5.6 orders of magnitude purely through units. We therefore report the *efficiency ratio* $\mathrm{RMSE}_{\mathrm{DP},j}/\mathrm{SE}_{\mathrm{OLS},j}$, which does not change when x_j or y is rescaled: a ratio r means the release is about as accurate as least squares on n/r^2 records (Figure 3). Once $\lambda = 0$ the ratios agree to within about $\pm 15\%$ across coefficients, with O-block coefficients about 10% higher than R-block ones.

Inference. With $\lambda = 0$, Wald intervals built from the first-order variance (sampling plus mechanism) covered the true slopes 93.5–95.9% of the time for $n \leq 5 \times 10^4$. With the ridge on, coverage of $\boldsymbol{\beta}$ collapsed for the large coefficients, which is what the bias-aware interval of

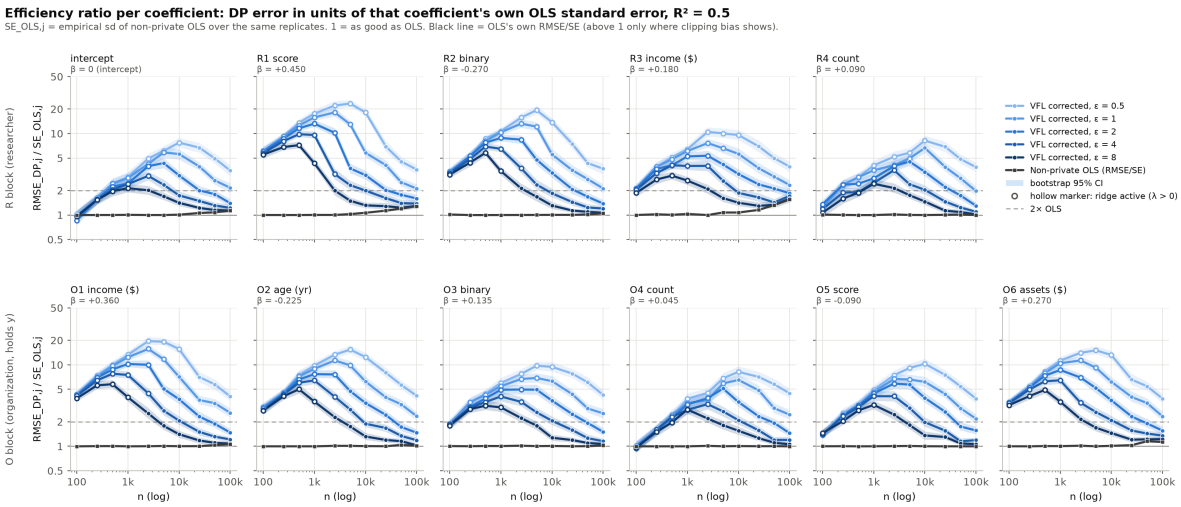


Figure 3: Efficiency ratio $\mathrm{RMSE}_{\mathrm{DP},j}/\mathrm{SE}_{\mathrm{OLS},j}$ per coefficient against n, $R^2 = 0.5$, earlier mechanism. Top row: R's block; bottom row: O's block.

Section 9.4 addresses. At large n coverage of the generating coefficients declines for least squares and the private estimate alike, because clipping moves the target; against the clipped-population target the same intervals cover 93–96%.

13.6 Does the bias correction matter? (earlier mechanism)

Not as a bias correction (Figure 4). In a single release the mechanism bias of Theorem 10.2 is at most 0.75% of the mean squared error, and the largest coefficient bias is at most 0.14 of its standard deviation. When σ is large the ridge is on, and 72–90% of the error against the true coefficients is deterministic shrinkage (Proposition 10.1), which the correction leaves alone by design. Against $\boldsymbol{\theta}_\lambda$ the correction lowers the mean squared error by 7–12% when the ridge is on, acting as a variance reduction, and by under 0.5% once $\vartheta_0 > 10$. Against the sample least-squares estimate or the true coefficients, with a moderate ridge on, it costs 0.3–4.3%. The bias itself matters only if many releases of the same data are averaged: at least about 130, growing like $23\vartheta_0^2$. On real data the correction changed test MSE by between -3.7% and $+2.3\%$. This is why v3.1 leaves it off by default.

[**Check 24:** Regenerate the figures with the current notation (theta-0 instead of rho-0) when they are rerun.]

13.7 Benchmark against AdaSSP (earlier mechanism)

AdaSSP [30] perturbs the same sufficient statistics and solves a ridge whose penalty adapts to a private estimate of $\lambda_{\min}(\mathbf{X}^\top\mathbf{X})$. It assumes one curator holding the joined table, which our setting rules out, so the comparison measures the utility cost of never pooling the data.

Setup. We use the 29 UCI datasets [11] of Wang's published evaluation, with his preprocessing (z-scored features, rows scaled to unit norm, $y/\max|y|$), 10-fold cross-validation, test MSE, $\delta = 10^{-6}$ and 50 noise draws per fold. Our re-implementation of Wang's procedures reproduces his published test MSE (median ratio 1.000 over 29 datasets and 10 values of ε; range 0.75–1.46).

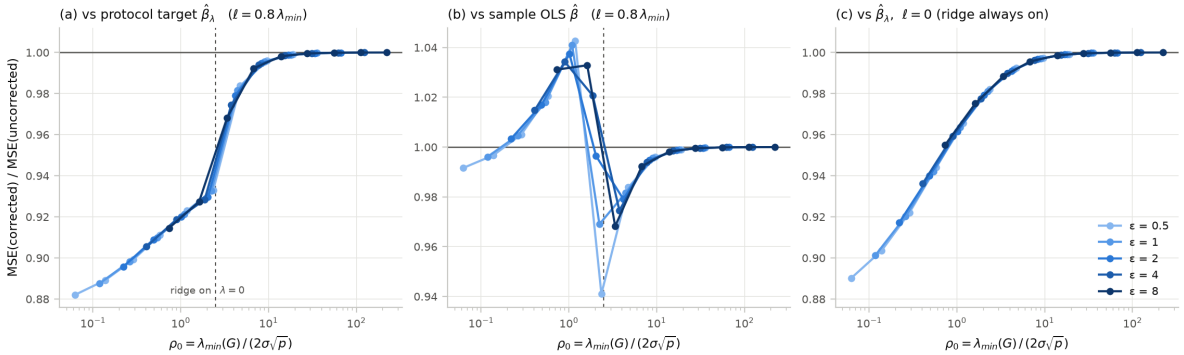


Figure 4: Mean squared error of the bias-corrected estimate divided by that of the uncorrected one, against $\vartheta_0 = \lambda_{\min}(\mathbf{G})/(2\sigma\sqrt{p})$ (labelled ρ_0 in the figure), one line per ε, earlier mechanism. (a) Against $\boldsymbol{\theta}_\lambda$; (b) against the sample least-squares estimate; (c) with $\ell = 0$, so that the ridge never switches off. Values below 1 favour the correction.

R holds the first $\lfloor d/2 \rfloor$ features and O the rest and y; every training row counts as matched, so no linkage is simulated. The preprocessing gives $\|\mathbf{x}_R\|, \|\mathbf{x}_O\|, |y| \le 1$, so $B_R = B_O = B_y = 1$ and $\Delta = \sqrt{10}$. No public lower bound on $\lambda_{\min}$ is available for these designs, so $\ell = 0$. We compare four private methods:

- *AdaSSP (published):* Wang's algorithm and constants. It protects the whole record $(\mathbf{x}, y)$, a stronger guarantee than ours.
- *AdaSSP (matched):* AdaSSP's adaptive ridge under our adjacency and accountant, with $\mathbf{A}$ exact; $\varepsilon/3$ is spent on $\lambda_{\min}$. It is an idealized reference, since its private $\lambda_{\min}$ step needs $\mathbf{X}^\top\mathbf{X}$ in the clear.
- *VFL, earlier mechanism:* the single release with the fixed ridge and the correction (15); a draw that fails the gate releases nothing ($\tilde{\boldsymbol{\theta}} = 0$).
- *VFL, uncorrected:* the same release and ridge without the correction.

Every method sees the same folds, and methods are compared by paired tests over folds (95% CI of the mean difference, t with 9 degrees of freedom).

[**Check 25:** Real-data preprocessing: Wang's pipeline normalizes whole rows across both parties' features, which no party can do in the vertical setting, and scales y by the data maximum. Redo the comparison under v3.1 with per-party preprocessing (block normalization) and bounds from public reference data, against AdaSSP with Wang's joint normalization.]

Results. At small ε the private VFL estimate is ahead of the published AdaSSP, which spends a third of its budget on $\lambda_{\min}$ (better on 21 datasets and worse on none at $\varepsilon = 0.01$). From $\varepsilon \approx 0.5$ AdaSSP is ahead: at $\varepsilon = 1$ the VFL test MSE is a median 4% higher (range -6% to $+31\%$), and the gap shrinks with n. Under the same guarantee the VFL estimate beats AdaSSP's adaptive ridge on 27 of 29 datasets at $\varepsilon \le 1$.

The published column should be read with one caveat. At $\varepsilon = 1$ AdaSSP adds noise of standard deviation 11.9 to each diagonal entry of $\mathbf{X}^\top\mathbf{X}$ and 8.4 to each off-diagonal entry, against our $\sigma = 13.4$, although its guarantee covers the whole record and it splits ε three ways.

[**Check 26:** Reconcile AdaSSP's published noise constants with ours (possibly a looser adjacency in Wang's

Table 17: Test MSE at $\varepsilon = 1$, as a ratio to non-private ridge regression on the same folds (earlier mechanism). Bold: lowest private method.

Dataset	n	d	Non-private test MSE	AdaSSP (published)	AdaSSP (matched)	VFL, earlier	VFL, uncorrected
housing	506	13	0.0396	**1.80**	2.07	1.97	1.95
wine	1,599	11	0.0201	**1.68**	1.95	1.84	1.83
airfoil	1,503	5	0.0533	**1.09**	1.17	1.16	1.16
concrete	1,030	8	0.0444	**1.48**	1.74	1.69	1.68
bike	17,379	17	0.0279	**1.03**	1.07	1.06	1.06
elevators	16,599	18	0.00868	**1.51**	1.78	1.71	1.70
pol	15,000	26	0.135	**1.04**	1.08	1.06	1.06
kin40k	40,000	8	0.0632	**1.00**	1.00	1.00	1.00
protein	45,730	9	0.119	**1.01**	1.01	1.01	1.01

Table 18: Paired comparisons over all 29 datasets (earlier mechanism): number of datasets where it has significantly lower / higher test MSE (95% CI over folds). The remaining datasets show no significant difference.

VFL (earlier), compared with	$\varepsilon = 0.01$	$\varepsilon = 0.1$	$\varepsilon = 1$	$\varepsilon = 10$
vs AdaSSP (published)	21 / 0	14 / 8	6 / 19	1 / 21
vs AdaSSP (matched)	27 / 0	27 / 0	27 / 0	21 / 3
vs Uncorrected VFL	23 / 6	14 / 8	6 / 19	3 / 19

calibration constant); until then the published column may overstate AdaSSP's advantage.]

13.8 Intersection size on real data (earlier mechanism)

Figure 6 repeats the experiment of Figure 2 on the four datasets with at least 15,000 rows and a real linear signal (bike, elevators, pol and protein). For each fold, the training folds are shuffled once and the first n rows form the matched set, nested across n; the held-out fold is the test set. Table 19 gives the n at which each method comes within 10% of non-private test MSE.

Doubling ε roughly halves the records needed for prediction. Over this range VFL needs 1.4–2.1 times the records AdaSSP needs. On bike, pol and protein, prediction comes within 10% of non-private at 1.4k–11k matched records for $\varepsilon \in \{1, 2\}$; elevators never does at the sizes available.

13.9 Where the utility goes

Prediction is not interpretation. Test MSE includes the irreducible noise, so it is forgiving. The coefficients themselves, which are what a researcher asking "do my variables predict this outcome?" needs, take about ten times more data, and intervals for $\boldsymbol{\beta}$ are informative only once the ridge is off or a public bound on $\|\boldsymbol{\beta}\|$ is available.

The gap to AdaSSP is the noise level, not the ridge rule. Replacing the fixed ridge by an oracle one changed test MSE by at most 1.2% at $\varepsilon = 1$; removing the ridge was far worse (27 times non-private on housing at $\varepsilon = 1$). What separates the methods is σ relative to the data.

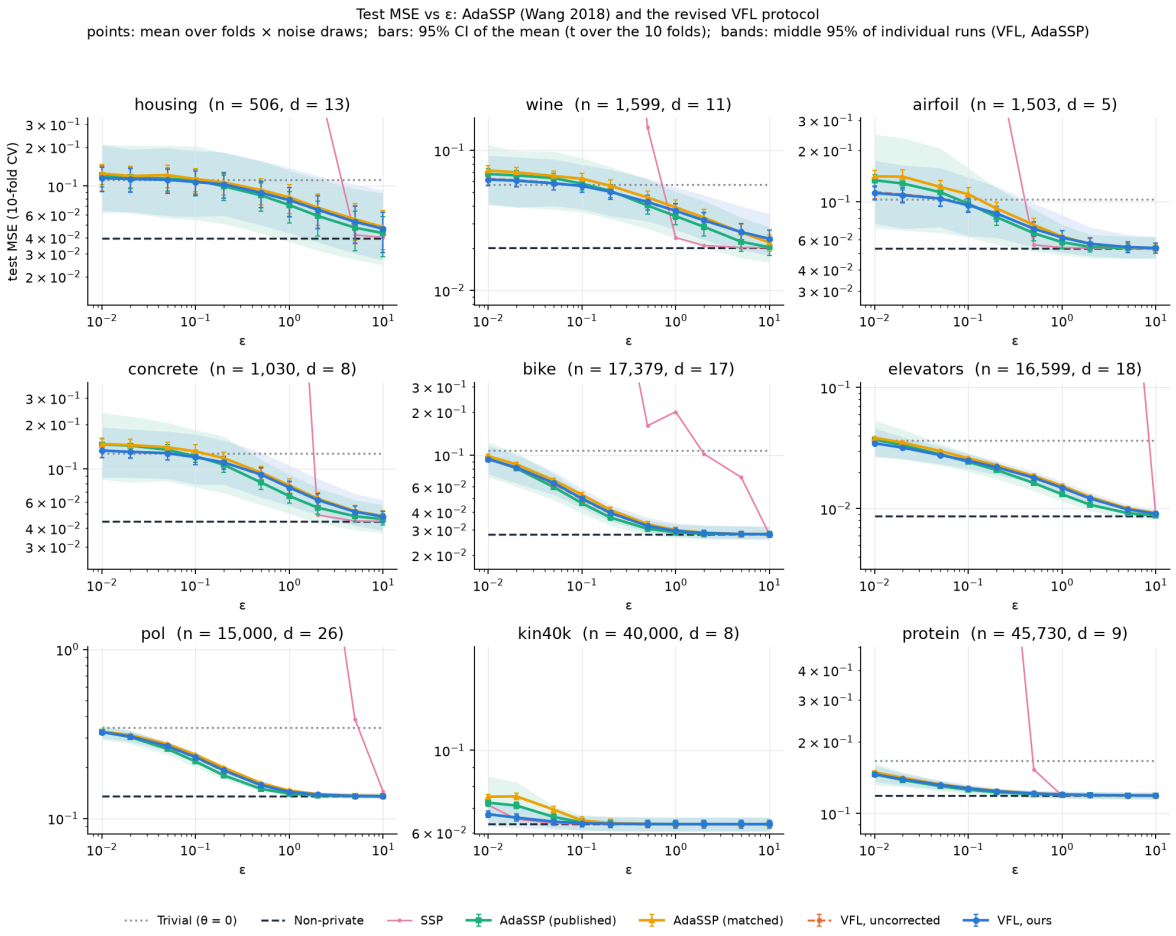


Figure 5: Test MSE against ε on nine of the datasets, earlier mechanism. Points: mean over 10 folds and 50 noise draws. Error bars: 95% CI of the mean (t over folds). Bands: middle 95% of individual runs.

Wang's preprocessing normalizes the whole row, so every record fills its bound. No party can do that in the vertical setting, but the per-party counterpart, block normalization (Section 6.1), recovers most of the gain: on protein it lowers Δ from 87.3 to 8.4 and brings private test error within 2–6% of non-private (Table 15).

How well the data fill their bounds matters as much as n. The noise is sized for the committed bounds, so headroom the data do not use is lost signal. After scaling by $\max|y|$, the typical $|y|$ is only 0.19 of its bound on elevators and 0.24 on wine, the two worst performers relative to their size. Choosing bounds that typical records come close to, and units or transforms that make this possible (for example log income), is part of using the protocol well.

Conditioning decides when the coefficients become usable. The ridge is off from about $(g^\star + r_1)/\ell$ matched records (Section 9.5). On protein a near-collinearity involving a single feature keeps it on until 691k records with block normalization and 13M without, at $\varepsilon = 1$;

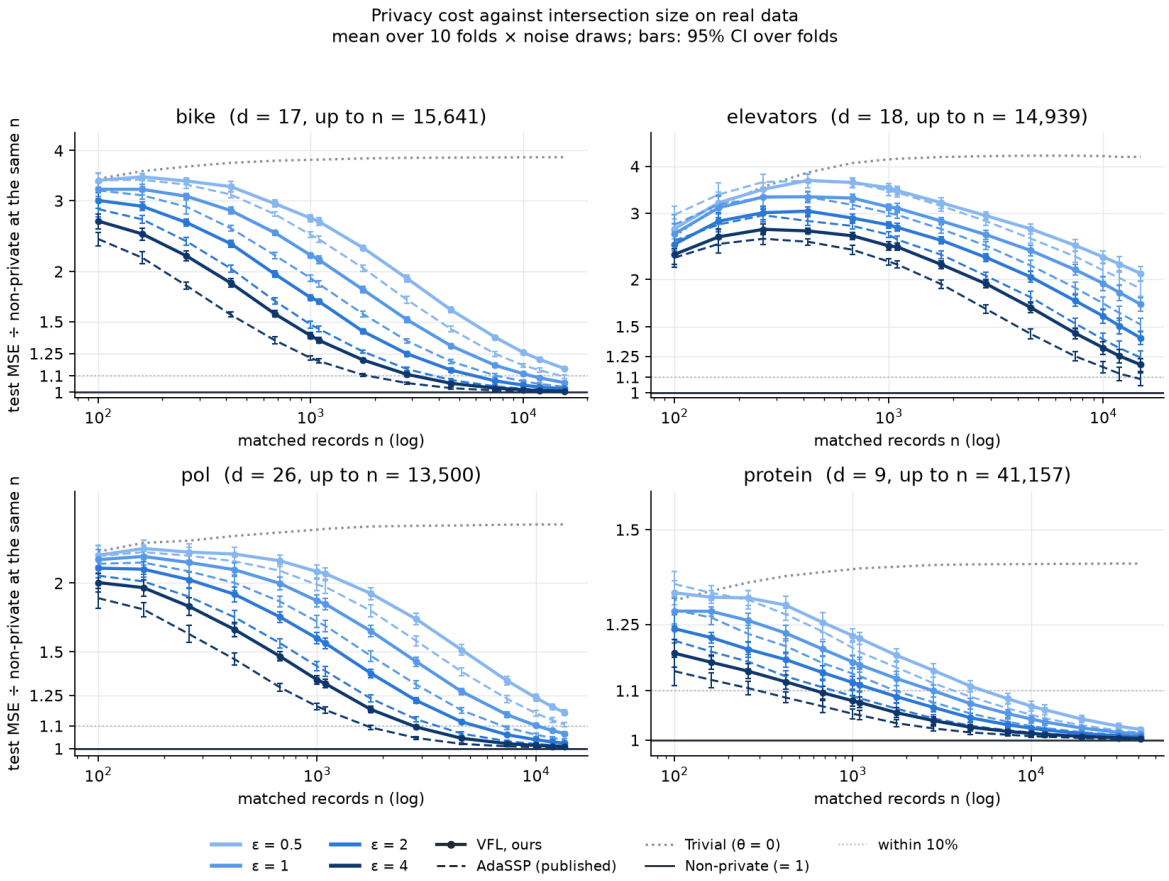


Figure 6: Test MSE divided by non-private test MSE at the same n, against the number of matched records, earlier mechanism. Solid: VFL; dashed: AdaSSP (published); colour: ε. Error bars: 95% CI over folds.

removing that feature, a decision that public reference data can inform, lowers the threshold to 102k.

Budget. Every release, including a refused one, is charged to O's ledger. Model search and exploratory re-runs spend the budget quickly; the protocol suits a pre-specified model per research question.

14 Cost

The measured runs use the chain of Section 12, with $\Omega_U = q_2$ and $\kappa = 40$; the operations, and so the timings, are the same at the evaluation settings.

Measured and extrapolated cost. On one core of a 2.8 GHz Xeon, a full run took 22–43 s end to end for $\bar{n}_O$ from 1,024 to 65,536 ($n_R^{\max} = 2{,}000$, $p = 11$). A cost model fitted to these runs, within about 10% of them, extrapolates O's CPU time to about 52 s, 99 s and 260 s for $\bar{n}_O = 10^5$, 3×10^5 and 10^6 (76 s on four cores at 10^6), with 88–422 MB of traffic, most of it R's

Table 19: Matched records needed to come within 10% of non-private test MSE: VFL (earlier mechanism), with AdaSSP (published) in brackets. ">" means not reached at the sizes available.

Dataset	n available	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$
bike	15,641	>15k (15k)	11k (7.5k)	5.9k (3.8k)	3.1k (1.8k)
elevators	14,939	>14k (>14k)	>14k (>14k)	>14k (>14k)	>14k (14k)
pol	13,500	>13k (>13k)	10k (7.0k)	5.3k (3.5k)	2.8k (1.7k)
protein	41,157	5.4k (3.7k)	2.8k (1.7k)	1.4k (784)	619 (295)

Table 20: Cost per run, with $m = \lceil \bar{n}_O / N_s \rceil$ chunks and $|\mathcal{E}|$ released entries. O's computation is $\Omega(\bar{n}_O)$, which is inherent for oblivious selection; the public scope Φ is the main lever on it.

Step	Communication	O's computation	Rounds
Link	$n_R^{\max}$ OPRF calls; PSI query and response (sublinear in $\bar{n}_O$)	per-run preprocessing $O(\bar{n}_O)$; per-query evaluation	2–3
Upload	$(d_R + 2)\, m$ seeded ciphertexts (half size)	$(d_R + 2)\, m$ key conversions	1
Aggregate	–	$\lvert\mathcal{E}\rvert\, m$ ciphertext–plaintext products; $\lvert\mathcal{E}\rvert \log_2 N_s$ rotations; $\lvert\mathcal{E}\rvert$ mask products	–
Release (×2 or 3)	2 ring elements each	1 re-randomization, 1 partial decryption, N exact samples	2–3

upload. At 10^6 rows the largest costs are O's OPRF evaluation of its whole register (fresh key every run), the encoding of its columns in the run's shuffled order, and the PSI interpolation; the CKKS aggregation itself is about 100 s on one core and parallelizes almost perfectly.

[**Check 27:** Run the staged benchmarks at 100k, 300k and 1M rows once the PSI item length and the datasets are decided.]

15 Extensions and limitations

A deviating researcher. v3.1 makes no claim against a malicious R. Making output privacy hold against one requires (i) zero-knowledge proofs at upload that the ciphertexts are well formed with short errors, that $b_t \in \{0, 1\}$, $\|\dot{\mathbf{x}}_t\| \le B_R$, $(1 - b_t)\dot{\mathbf{x}}_t = \mathbf{0}$ and $\sum_t b_t \le n_R^{\max}$, together with a coarse norm bound that rules out wraparound (candidates include lattice-based and VOLE-based proofs); (ii) key generation with R committing first, plus proofs that its shares are short; and (iii) OPRF inputs bound to a certified cohort. A homomorphic check that relies on R's own decryption share cannot do this job.

Remark 15.1 (Hiding membership)**.** Replacing labeled PSI with circuit-PSI (for example VOLE-PSI with payloads [27]) gives both parties shares of $\mathbf{b}$ and $\dot{\mathbf{X}}_R$ in τ-order, so R no longer learns I. The aggregation would move to BFV or use wide statistical masking, because shares modulo 2^ℓ do not reduce inside CKKS. $\mathbf{A}$ must then be computed under encryption and noised, and the adjacency extends to changing a record's match status, which raises the sensitivity.

Further limitations.

- Exact-identifier linkage only.
- Linear models only; generalized linear models via approximate sufficient statistics are future work.
- The variance is model-based. A sandwich variance would need a third release of $\sum_i r_i^2 \mathbf{x}_i \mathbf{x}_i^\top$, a ciphertext–ciphertext computation with its own sensitivity.
- No verifiability of O's computation.

16 Summary and open items

Protocol v3.1 links records with an OPRF and labeled PSI, aggregates under two-party threshold CKKS, and makes each release a threshold decryption in which O's share carries the DP noise and floods the encryption error. O never decrypts, R never sees a noiseless statistic that depends on O's data, and the budget is enforced by O's control of its share. Two releases separate the choice of the ridge from the estimate, which makes the bias theory and the intervals valid. The protocol and its guarantees are stated for any admissible parameter vector (Section 12); Calibrate derives the flooding parameter, the scale and the noise from it, and the ledger composes runs, fixed or adaptive, under a per-register cap.

Remaining obligations.

O1. The worst-case error bound: proven for the circuit as written, as an explicit function of the parameters ($B_{\text{tot}} \approx 2^{25.3}$ at the proofs' example); proofs for the restructured circuit are open.
O2. Smudging for the discrete Gaussian, for every $\sigma_c \geq 2^{20}$ (proven), and a concrete constant-time exact sampler (open).
O3. Simulation security of threshold key generation and rotation keys against a semi-honest R (reduced to RLWE with circular security).
O4. The two-stage rule, the RSS correction and the coverage of both intervals, checked by simulation (Section 13.3) and still to be checked on real data under v3.1.
O5. PSI parameters with explicit failure probabilities (correctness only, not charged to δ), an item length that meets them at $\bar{n}_O = 10^6$, and a worst-case bound B_ψ on the evaluation noise of the PSI replies, which SanitizeBFV floods.

Other open items. A written proof of the adaptive hybrid in Proposition 7.2; a sharper treatment of the smudging step (Remark 7.4); no-wraparound constants at the earlier levels; precise modulus-chain sizes; the free-set boundary for $\mathbf{y}^\top \mathbf{y}$ and a closed form for $d_O = 1$; explicit constants for the bootstrap RSS correction; a remedy for heteroskedasticity when clipping binds; and sixteen citation details flagged for verification in the proofs.

A Proof of Theorem 10.1

Throughout, condition on the sample so that $\mathbf{G}$, $\mathbf{c}$ and $\hat{\boldsymbol{\beta}} = \mathbf{G}^{-1}\mathbf{c}$ are fixed, and write $\mathbf{P} = \mathbf{G}^{-1}$ and $\mathbf{H} = \mathbf{PE}$.

Step 1: Neumann expansion on $\mathcal{E}_\eta$. On $\mathcal{E}_\eta$ the spectral radius of $\mathbf{H}$ is at most $\eta < 1$ (Remark 10.2), so

$$\tilde{\boldsymbol{\beta}} = (\mathbf{G}+\mathbf{E})^{-1}(\mathbf{c}+\mathbf{f}) = (\mathbf{I}+\mathbf{H})^{-1}\mathbf{P}(\mathbf{c}+\mathbf{f}) = \big(\mathbf{I}-\mathbf{H}+\mathbf{H}^2-\mathbf{H}^3+\mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}\big)\big(\hat{\boldsymbol{\beta}}+\mathbf{P}\mathbf{f}\big),$$

and $\|\mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}\|$ is bounded by a constant times $\|\mathbf{E}\|_{\mathrm{op}}^4/(1-\eta)$ in the $\mathbf{G}$-norm.

Step 2: odd terms and the moment noise vanish. $\mathcal{E}_\eta$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$, $\mathbf{E}$ is symmetric about zero, and $\mathbf{f}$ is independent of $\mathbf{E}$ and symmetric about zero. Hence $\mathbb{E}[\mathbf{H}^m \mid \mathcal{E}_\eta] = 0$ for odd m, and every term containing $\mathbf{P}\mathbf{f}$ has zero conditional expectation. Therefore

$$\mathbb{E}[\tilde{\boldsymbol{\beta}} \mid \mathcal{E}_\eta] = \hat{\boldsymbol{\beta}} + \mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta]\,\hat{\boldsymbol{\beta}} + O(\sigma^4).$$

Step 3: removing the conditioning.

$$\mathbb{E}[\mathbf{H}^2 \mid \mathcal{E}_\eta] - \mathbb{E}[\mathbf{H}^2] = \big(\mathbb{E}[\mathbf{H}^2\mathbf{1}_{\mathcal{E}_\eta}] - \mathbb{E}[\mathbf{H}^2]\Pr(\mathcal{E}_\eta)\big)/\Pr(\mathcal{E}_\eta),$$

whose norm is at most $2\big(\mathbb{E}\|\mathbf{H}\|^4\big)^{1/2}\Pr(\mathcal{E}_\eta^c)^{1/2}/\Pr(\mathcal{E}_\eta) = O\big(\sigma^2 e^{-p(\eta\vartheta_0-1)^2/2}\big)$ by Cauchy–Schwarz and Remark 10.2.

Step 4: the second-moment contraction. Write $\mathbf{p} = (P_{11},\dots,P_{pp})^\top$ for the diagonal of $\mathbf{P}$ and $\mathbf{X} = (\chi_{ij})$.

Lemma A.1. *For any fixed symmetric $\mathbf{P}$, with $\circ$ the Hadamard product, $\mathbb{E}[\mathbf{EPE}] = \sigma^2\big(\mathbf{X}\circ\mathbf{P} + \mathrm{diag}(\mathbf{Xp}) - \Pi_O\,\mathrm{diag}(\mathbf{p})\big)$.*

Proof. Entrywise, $\mathbb{E}[(\mathbf{EPE})_{ab}] = \sum_{i,j} P_{ij}\mathbb{E}[E_{ai}E_{jb}]$, and (11) gives $\mathbb{E}[E_{ai}E_{jb}] = \sigma^2\chi_{ai}(\delta_{aj}\delta_{ib} + \delta_{ab}\delta_{ij} - \delta_{ai}\delta_{aj}\delta_{ab})$. Summing against the three terms gives $\sigma^2\chi_{ab}P_{ab}$, $\sigma^2\delta_{ab}\sum_i \chi_{ai}P_{ii}$, and $-\sigma^2\delta_{ab}\chi_{aa}P_{aa}$. With $\chi_{aa} = \mathbf{1}[a \in O]$ these are the (a,b) entries of $\mathbf{X}\circ\mathbf{P}$, $\mathrm{diag}(\mathbf{Xp})$ and $-\Pi_O\,\mathrm{diag}(\mathbf{p})$. □

Writing $\chi_{ij} = 1 - (\Pi_R)_{ii}(\Pi_R)_{jj}$ turns the first two terms into projector form: $\mathbf{X}\circ\mathbf{P} = \mathbf{P} - \Pi_R\mathbf{P}\Pi_R$ and $\mathrm{diag}(\mathbf{Xp}) = \mathrm{tr}(\mathbf{P})\mathbf{I} - \mathrm{tr}(\Pi_R\mathbf{P})\Pi_R$. Left-multiplying by $\mathbf{P}$ and using $\Pi_O\,\mathrm{diag}(\mathbf{p}) = \mathrm{diag}(\mathbf{P})\Pi_O$,

$$\mathbb{E}[\mathbf{PEPE}] = \sigma^2\big[\mathbf{P}^2 - \mathbf{P}\Pi_R\mathbf{P}\Pi_R + \mathrm{tr}(\mathbf{P})\mathbf{P} - \mathrm{tr}(\Pi_R\mathbf{P})\mathbf{P}\Pi_R - \mathbf{P}\,\mathrm{diag}(\mathbf{P})\Pi_O\big].$$

Step 5: regrouping. Substituting $\Pi_O = \mathbf{I} - \Pi_R$, the first four terms collapse to $(\mathbf{P}^2 + \mathrm{tr}(\mathbf{P})\mathbf{P})\Pi_O + (\mathbf{P}\Pi_O\mathbf{P} + \mathrm{tr}(\Pi_O\mathbf{P})\mathbf{P})\Pi_R$, which gives $\mathbb{E}[\mathbf{H}^2] = \sigma^2 M(\mathbf{P})$. Combining with Steps 2 and 3 proves the theorem. □

Degenerate check. With $\Pi_O = \mathbf{I}$ and $\Pi_R = \mathbf{0}$ the formula gives $\sigma^2(\mathbf{P}^2 + \mathrm{tr}(\mathbf{P})\mathbf{P} - \mathbf{P}\,\mathrm{diag}(\mathbf{P}))$; for $p = 1$ this is $\sigma^2 a^{-2}\hat{\boldsymbol{\beta}}$, matching a direct expansion of $c/(a+e)$ with $\mathrm{Var}(e) = \sigma^2$.

Extension to ridge (Theorem 10.2). Replacing $\mathbf{G}$ by $\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi$ with (λ, Ψ) independent of $(\mathbf{E}, \mathbf{f})$ leaves $\mathbf{E}$, its covariance and $\mathbf{f}$ untouched, and Steps 1–5 hold verbatim with $\mathbf{P} \mapsto \mathbf{P}_\lambda$. If λ depended on $\mathbf{E}$, Step 2 would fail: $\mathbf{P}_\lambda$ would no longer be fixed under the conditional expectation.

B Labeled PSI with an OPRF and record alignment

R must learn I and the alignment π, while O learns only public sizes. We use labeled PSI in the style of Chen, Laine and Rindal [7, 8, 10] on OPRF values. The construction runs under BFV with R's own key, whose exact arithmetic modulo t is needed for the zero test and label recovery. Algorithm 2 gives the steps; the toy example deck works them through with numbers.

Pre-processing at O (fresh every run). O pads its scope register to $\bar{n}_O$, draws a uniformly random permutation τ of its rows and a fresh OPRF key k. Each row j becomes the item $F_k(\mathrm{id}_j)$, mapped into $\mathbb{Z}_t$ so as to avoid the padding root $t-1$, with label $\tau(j)$ encrypted under a key derived from $F'_k(\mathrm{id}_j)$. Each item is hashed into all h candidate bins; each bin is split into α partitions of at most D items, where D and α are public functions of $(\bar{n}_O, h, N_{\mathrm{bins}}, d_{\mathrm{cap}})$ chosen so that overflow has probability at most δ_{psi}. Every partition's membership polynomial is padded with the root $t-1$ to degree exactly D, and its label polynomial interpolates (item, encrypted label) pairs.

Query at R. R pads its identifiers to $n_R^{\max}$, obtains F_k and F'_k of each through the OPRF, cuckoo-hashes [25] the items into N_{bins} bins with the same h hash functions (at most one item per bin), and sends BFV encryptions of the slot-wise powers $\mathbf{w}^{2^i}$, $i < W$, with $W = \lceil \log_2 D \rceil + 1$.

Evaluation and reply at O. O forms the encryptions of $\mathbf{w}^d$, $d \le D$, as products of window ciphertexts, evaluates the membership and label polynomials of each partition as plaintext-coefficient linear combinations, masks with $\mathbf{r}_1, \mathbf{r}_2$ uniform in $(\mathbb{Z}_t^*)^{N_{\mathrm{bins}}}$ from a CSPRNG, and returns

$$\textsc{SanitizeBFV}\big(\mathbf{r}_1 \odot P(\mathbf{w})\big), \qquad \textsc{SanitizeBFV}\big(\mathbf{r}_2 \odot P(\mathbf{w}) + L(\mathbf{w})\big).$$

The first is 0 in a bin iff R's item there is in O's set, and the second is then that item's encrypted label; otherwise both are uniform. The mask $\mathbf{r}_1$ hides $P(\mathbf{w})$ at non-matches, which would otherwise carry information about O's items. SanitizeBFV and its parameters $(\kappa_\psi, B_\psi, q')$ are discussed under G4 in Section 8.

Recovery at R. For each occupied bin and partition, R opens the label under F'_k of its item when the flag is zero, which gives $\pi(i) = \tau(j)$.

Remark B.1 (Collision probability)**.** A false match requires two distinct identifiers to share an item value inside a common bin. The expected number is about $n_R^{\max}\bar{n}_O h/(N_{\mathrm{bins}} t)$, so $t \ge n_R^{\max}\bar{n}_O h/(N_{\mathrm{bins}}\delta_{\mathrm{psi}})$ (a prime with $t \equiv 1 \bmod 2N_B$ for batching) keeps it below δ_{psi}. The padding root must be $t-1$ for the t actually used. The evaluation setting $t \approx 2^{32}$ does not meet this constraint for $\delta_{\mathrm{psi}} = 2^{-40}$: with $n_R^{\max} = 2{,}000$, $h = 3$ and $N_{\mathrm{bins}} = 2^{14}$, the expected number of false matches per run is 3.5×10^{-7} at $\bar{n}_O = 4{,}096$ and 8.5×10^{-5} at 10^6. R rejects ambiguous and out-of-range labels, but an in-range false match goes undetected, so longer items are needed.

Remark B.2 (Pre-shuffle necessity)**.** Without τ, the positions $\pi(i)$ recovered by R would reflect O's original database ordering (e.g. by region or registration date), leaking attributes outside the DP guarantee. The fresh shuffle makes positions uniformly random. It must be applied by O inside the protocol, not assumed of the input files.

Remark B.3 (Linkage privacy)**.** O sees $n_R^{\max}$ uniformly random group elements in the OPRF and BFV ciphertexts under R's key in the PSI, so it learns neither I nor n_R.

C Status of the proofs

The companion proofs document (Part A: cryptography and DP; Part B: sensitivity and statistics) labels every step as exact, first order, asymptotic, numerical or open. Table 21 summarizes it.

Table 21: Status of the claims.

Claim	Status	Notes
Encoding sensitivity (Lemma 7.1)	proven	$\sqrt{2/N}$ and the $\sqrt{N}$ rounding term
Smudging (Lemma 7.2)	proven	explicit constants
Worst-case B_{tot} (Theorem 7.1)	proven, explicit	$\approx 2^{25.3}$ for the circuit as written; checked against OpenFHE
zCDP of the ideal mechanism; tight conversion	proven	$\rho_{\text{run}} = 0.01496$ at $\varepsilon = 1$, $\delta = 10^{-9}$
Output privacy (Theorem 7.2)	proven (hybrids)	RLWE in two forms; circular security for O's rotation-key shares
Input and cohort privacy (G1, G2)	proven	simulators, up to the cuckoo-hashing failure probability
Correctness (no wraparound)	proven	failure probability $\leq \delta_B + 2Ne^{-t_\eta^2/2}$ per release
Composition across runs (Proposition 7.2)	non-adaptive proven; adaptive argued	the adaptive hybrid is not yet written out in the proofs
G4 with SanitizeBFV	argued	needs a worst-case bound B_ψ on the BFV evaluation noise
Replace-one sensitivity (Proposition 7.1)	proven, exact for $d_O \geq 2$	regime 3 reduces to three cubics; $d_O = 1$ valid but conservative
When $\mathbf{y}^\top\mathbf{y}$ is free	partly proven	boundary between $2(r+o)$ and $3(r+o)$ numerical only
Operator-norm radius r_1	proven (one-sided)	about $1.55\times$ conservative
Two-stage (λ, Ψ)	proven	independent of the stage-2 noise; closed form for λ_O
Bound on $\|\boldsymbol{\beta}\|$	proven	for the projection coefficient, no model assumed
Coverage	partly first order	Wald remainder exact; plug-in variance first order; normality asymptotic
Bootstrap RSS correction	first order	constants not explicit

[**Check 28:** The proofs document and the v3.1 specification are not yet in the repository; add them (for example under docs/) so that this paper can cite them.]

References

[1] M. Albrecht, M. Chase, H. Chen, J. Ding, S. Goldwasser, S. Gorbunov, S. Halevi, J. Hoffstein, K. Laine, K. Lauter, S. Lokam, D. Micciancio, D. Moody, T. Morrison, A. Sahai,

and V. Vaikuntanathan, "Homomorphic encryption security standard," HomomorphicEncryption.org, Toronto, Canada, November 2018.

[2] T. B. Armstrong, M. Kolesár, and S. Kwon, "Bias-aware inference in regularized regression models," arXiv:2012.14823, 2020.

[3] G. Asharov, A. Jain, A. López-Alt, E. Tromer, V. Vaikuntanathan, and D. Wichs, "Multiparty computation with low communication, computation and interaction via threshold FHE," in *Proc. EUROCRYPT*, 2012.

[4] B. Balle and Y.-X. Wang, "Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising," in *Proc. ICML*, 2018, pp. 394–403.

[5] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Proc. TCC*, 2016.

[6] C. L. Canonne, G. Kamath, and T. Steinke, "The discrete Gaussian for differential privacy," in *Proc. NeurIPS*, 2020, pp. 15676–15688.

[7] H. Chen, K. Laine, and P. Rindal, "Fast private set intersection from homomorphic encryption," in *Proc. ACM CCS*, 2017.

[8] H. Chen, Z. Huang, K. Laine, and P. Rindal, "Labeled PSI from fully homomorphic encryption with malicious security," in *Proc. ACM CCS*, 2018.

[9] J.-H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Proc. ASIACRYPT*, 2017, pp. 409–437.

[10] K. Cong, R. Cruz Moreno, M. B. da Gama, W. Dai, I. Iliashenko, K. Laine, and M. Rosenberg, "Labeled PSI from homomorphic encryption with reduced computation and communication," in *Proc. ACM CCS*, 2021.

[11] D. Dua and C. Graff, *UCI Machine Learning Repository*. University of California, Irvine, School of Information and Computer Sciences, 2019.

[12] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Now Publishers, 2014.

[13] V. Feldman and T. Zrnic, "Individual privacy accounting via a Rényi filter," in *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021. arXiv:2008.11193.

[14] M. J. Freedman, K. Nissim, and B. Pinkas, "Efficient private matching and set intersection," in *Proc. EUROCRYPT*, 2004, pp. 1–19.

[15] D. Froelicher, J. R. Troncoso-Pastoriza, J. S. Sousa, and J.-P. Hubaux, "Drynx: Decentralized, secure, verifiable system for statistical queries and machine learning on distributed datasets," *IEEE Trans. Inf. Forensics Security*, 2020.

[16] D. Froelicher, J. R. Troncoso-Pastoriza, J. L. Raisaro, M. A. Cuendet, J. S. Sousa, H. Cho, B. Berger, J. Fellay, and J.-P. Hubaux, "Truly privacy-preserving federated analytics for precision medicine with multiparty homomorphic encryption," *Nature Communications*, vol. 12, 2021.

[17] A. Gascón, P. Schoppmann, B. Balle, M. Raykova, J. Doerner, S. Zahur, and D. Evans, "Privacy-preserving distributed linear regression on high-dimensional data," *Proc. Privacy Enhancing Technologies*, 2017.

[18] S. Hardy, W. Henecka, H. Ivey-Law, R. Nock, G. Patrini, G. Smith, and B. Thorne, "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," arXiv:1711.10677, 2017.

[19] S. Jarecki, A. Kiayias, and H. Krawczyk, "Round-optimal password-protected secret sharing and T-PAKE in the password-only model," in *Proc. ASIACRYPT*, 2014.

[20] B. Li and D. Micciancio, "On the security of homomorphic encryption on approximate numbers," in *Proc. EUROCRYPT*, 2021.

[21] B. Li, D. Micciancio, M. Schultz, and J. Sorrell, "Securing approximate homomorphic encryption using differential privacy," in *Proc. CRYPTO*, 2022.

[22] I. Mironov, O. Pandey, O. Reingold, and S. Vadhan, "Computational differential privacy," in *Proc. CRYPTO*, 2009.

[23] I. Mironov, "On significance of the least significant bits for differential privacy," in *Proc. ACM CCS*, 2012, pp. 650–661.

[24] C. Mouchet, J. Troncoso-Pastoriza, J.-P. Bossuat, and J.-P. Hubaux, "Multiparty homomorphic encryption from ring-learning-with-errors," *Proc. Privacy Enhancing Technologies*, 2021.

[25] R. Pagh and F. F. Rodler, "Cuckoo hashing," *J. Algorithms*, vol. 51, no. 2, 2004.

[26] A. Davidson, A. Faz-Hernández, N. Sullivan, and C. A. Wood, "Oblivious pseudorandom functions (OPRFs) using prime-order groups," RFC 9497, 2023.

[27] P. Rindal and P. Schoppmann, "VOLE-PSI: fast OPRF and circuit-PSI from vector-OLE," in *Proc. EUROCRYPT*, 2021.

[28] R. Rogers, A. Roth, J. Ullman, and S. Vadhan, "Privacy odometers and filters: Pay-as-you-go composition," in *Advances in Neural Information Processing Systems 29 (NIPS)*, 2016, pp. 1921–1929.

[29] O. Sheffet, "Differentially private ordinary least squares," in *Proc. ICML*, 2017.

[30] Y.-X. Wang, "Revisiting differentially private linear regression: optimal and adaptive prediction & estimation in unbounded domain," in *Proc. UAI*, 2018. Code and data: `https://github.com/yuxiangw/optimal_dp_linear_regression`.

[31] J. Whitehouse, A. Ramdas, R. Rogers, and Z. S. Wu, "Fully-adaptive composition in differential privacy," in *Proc. ICML*, PMLR vol. 202, 2023, pp. 36990–37007.

[32] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, 2019.